# Statistics for Economics and Management

## Workbook

### Spring Semester 2025/2026

# Contents


**1 Review on random variables** **2**

**2 Sampling distributions** **9**

**3 Point estimation** **16**

**4 Confidence intervals** **19**

**5 Hypothesis testing** **28**

**6 Linear regression** **36**

**7 Introduction to categorical data analysis** **49**

**8 Solutions of the proposed exercises** **52**

# 1 Review on random variables

1. Using the definitions, derive the expected value and variance of a Bernoulli random variable.

2. For the next academic year, the UVP University will require all its students to purchase their own laptop to use in class. To assess the reaction of UVP students to this new policy, all students of the university were surveyed. To the question "What is your opinion about the proposed measure?" each student could answer "Agree", "Disagree", or "Indifferent". The results, in percentage, are presented in the following table which also distinguishes between undergraduate and master students.

| | Program | |
|---|---|---|
| Opinion | Undergraduate | Master |
| Agree | 20% | 30% |
| Disagree | 10% | 20% |
| Indifferent | 10% | 10% |

   Suppose that a student of UVP University was randomly chosen. What is the probability that this student answered "Disagree" in the following two cases?

   (a) Nothing more is known about the student.

   (b) It is known that it is a Master student.

3. (Newbold, p. 189)[1] A researcher suspects that the number of between-meal snacks ($Y$) eaten by students in a day during final examinations might depend on the number of tests ($X$) a student had to take on that day. The accompanying table shows joint probabilities, estimated from a survey.

| **Y\X** | **0** | **1** | **2** | **3** |
|---|---|---|---|---|
| **0** | 0.07 | 0.09 | 0.06 | 0.01 |
| **1** | 0.07 | 0.06 | 0.07 | 0.01 |
| **2** | 0.06 | 0.07 | 0.14 | 0.03 |
| **3** | 0.02 | 0.04 | 0.16 | 0.04 |

   (a) Find the probability distribution of $X$ and compute the mean number of tests taken by students on that day.

   (b) Find the probability distribution of $Y$ and, hence, the mean number of snacks eaten by students on that day.

   (c) Find and interpret the conditional probability distribution of $Y$, given that $X = 3$.

   (d) Find the covariance between $X$ and $Y$.

   (e) Are number of snacks and number of tests independent of each other?

4. Consider two random variables $X_1$ and $X_2$ with joint probabilities given in the following table:

| $\mathbf{X_2 \backslash X_1}$ | **-1** | **0** | **1** |
|---|---|---|---|
| **-1** | 0.2 | 0.1 | 0 |
| **0** | 0.1 | 0.2 | 0.1 |
| **1** | 0 | 0.1 | 0.2 |

   Compute the variances of $X_1$ and $X_2$, their covariance and correlation. What is the variance of $X_1 + X_2$?

[1] The pages refer to the 8th edition of: Newbold, P., Carlson, W. and Thorne, B. *Statistics for Business and Economics*

5. (Newbold, p. 239) The random variable $X$ has probability density function as follows:

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \\ 0, & \text{other } x \end{cases}$$

   (a) Graph the probability density function for $X$.
   (b) Show that the density has the properties of a proper probability density function.
   (c) Find the probability that $X$ takes a value between 0.5 and 1.5.

6. Using the definitions, derive the expected value and variance of a continuous random variable that is uniformly distributed over an interval $[a, b]$.

7. (Newbold, p. 203) The jurisdiction of a rescue team includes emergencies occurring on a stretch of river that is 4 miles long. Experience has shown that the distance along this stretch, measured in miles from its northernmost point, at which an emergency occurs can be represented by a uniformly distributed random variable over the range 0 to 4 miles. Then, if $X$ denotes the distance (in miles) of an emergency from the northernmost point of this stretch of river, its probability density function is as follows:

$$f(x) = \begin{cases} 0.25, & 0 < x < 4 \\ 0, & \text{other } x \end{cases}$$

   (a) Graph the probability density function.
   (b) Find and graph the cumulative distribution function.
   (c) Find the probability that a given emergency arises within 1 mile of the northernmost point of this stretch of river.
   (d) The rescue team's base is at the midpoint of this stretch of river. Find the probability that a given emergency arises more than 1.5 miles from this base.

8. (Newbold, p. 206) An author receives a contract from a publisher, according to which she is to be paid a fixed sum of \$10000 plus \$1.50 for each copy of her book sold. Her uncertainty about total sales of the book can be represented by a random variable with a mean of 30000 and a standard deviation of 8000. Find the mean and standard deviation of the total payments she will receive.

9. (Newbold, p. 206) A contractor submits a bid on a project for which research and development work needs to be done. It is estimated that the total cost of satisfying the project specifications will be \$20 million plus the cost of the further research and development work. The contractor views the cost of this additional work as a random variable with a mean of \$4 million and a standard deviation of \$1 million. The contractor wishes to submit a bid such that his expected profit will be 10% of his expected costs. What should be the bid? If this bid is accepted, what will be the standard deviation of the profit made by the project?

10. (Newbold, p. 237) Five inspectors are employed to check the quality of components produced on an assembly line. For each inspector the number of components that can be checked in a shift can be represented by a random variable with mean 120 and standard deviation 15. Let X represent the number of components checked by an inspector in a shift. Then the total number checked is 5X, which has a mean of 600 and a standard deviation of 75. What is wrong with this argument? Assuming that inspectors' performances are independent of one another, find the mean and standard deviation of the total number of components checked in a shift.

11. (Newbold, p. 239) Let $X_1$ and $X_2$ be a pair of random variables. Show that the covariance between the random variables $Y_1 = (X_1 + X_2)$ and $Y_2 = (X_1 - X_2)$ is 0 if and only if $X_1$ and $X_2$ have the same variance.

12. (Newbold, p. 237) Shirley Johnson, portfolio manager, has asked you to analyze a newly acquired portfolio to determine its mean value and variability. The portfolio consists of 50 shares of Xylophone Music and 40 shares of Yankee Workshop. Analysis of past history indicates that the share price of Xylophone Music has a mean of 25 and a variance of 121. A similar analysis indicates that Yankee has a mean share price of 40 with a variance of 225. Your best evidence indicates that the share prices have a correlation of 0.5.

    (a) Compute the mean and variance of the portfolio.

    (b) Suppose that the correlation between share prices was actually -0.5. Now what are the mean and variance of the portfolio?

13. In the final two weeks of a promotion campaign, a firm must decide who will be the operational supervisor in each of these two weeks. There are two candidates: Carla and Maria. On average, both Carla and Maria are expected to achieve the same sales per week. Assume that the standard deviation of weekly sales is also the same for both. If Carla supervises in one week, and Maria supervises in the other week, sales in each week are independent. However, if the same person supervises in both weeks, sales in each week are positively correlated since if one of the supervisors performs above (below) average in the first week, it is more likely that she will also perform above (below) average in the second week. This joint decision for the two weeks has to be made today, in advance of those two weeks, and cannot be changed afterwards.

    (a) In terms of expected total sales in the two weeks, what is the best choice of supervisor for each week? Statistically justify your answer.

    (b) If the firm is risk averse, what is the best choice of supervisors in each week? Provide the statistical details to justify your answer.

14. You are given 2 euros that must be spent on a "head=keep or tail=lose" coin flipping games. You can choose between two options:

    - Option A - betting all your 2 euros on one game (so that you either lose 2 euros or earn 2 euros, with equal probability);
    - Option B - spend 1 euro in one such game and another 1 euro on another such game (the results of these two games are independent).

    (a) Which of the two options A or B is better in terms of expected value?

    (b) And in terms of less risk (captured by a smaller variance)?

    (c) Discuss why this exercise provides an example of how financial risk can be reduced by diversifying the investments.

15. According to a financial market analyst the expected return for each thousand euros invested in two investment funds A and B are 40 and 120 euros respectively. According to the same analyst, the corresponding standard deviations are equal to 10 and 50, respectively, and the correlation between the two returns equals $-0.4$. Suppose you invest 15 thousand euros as follows: 5 thousand euros in A and 10 thousand euros in B.

    (a) What is the expected return of your total investment?

    (b) What is the standard deviation of the return of your total investment?

    (c) What is the expected return and standard deviation if you invest all the 15 thousand euros in A?

    (d) Using Excel and the Solver add-in, find the optimal portfolio composition (given a total amount of 15 thousand Euros to invest) for each of the following objectives:

        i. Maximize expected returns

ii. Minimize risk (standard deviation)
iii. Minimize risk assuring an expected return of no less than 600 euros (compare with previous results).
iv. Minimize risk assuring an expected return of no less than 1000 euros.

16. (Newbold, p. 217) Let the random variable $Z$ follow a standard normal distribution.

    (a) Find $P(Z < 1.20)$.
    (b) Find $P(Z > 1.33)$.
    (c) Find $P(Z > -1.70)$.
    (d) Find $P(Z > -1.00)$.
    (e) Find $P(1.20 < Z < 1.33)$.
    (f) Find $P(-1.70 < Z < 1.20)$.
    (g) Find $P(-1.70 < Z < -1.00)$.

17. (Newbold, p. 217) Let the random variable $Z$ follow a standard normal distribution.

    (a) The probability is 0.70 that $Z$ is less than what number?
    (b) The probability is 0.25 that $Z$ is less than what number?
    (c) The probability is 0.2 that $Z$ is greater than what number?
    (d) The probability is 0.6 that $Z$ is greater than what number?

18. (Newbold, p. 217) Let the random variable $X$ follow a normal distribution with $\mu = 50$ and $\sigma^2 = 64$.

    (a) Find the probability that $X$ is greater than 60.
    (b) Find the probability that $X$ is greater than 35 and less than 62.
    (c) Find the probability that $X$ is less than 55.
    (d) The probability is 0.2 that $X$ is greater than what number?
    (e) The probability is 0.05 that $X$ is in the symmetric interval about the mean between which two numbers?

19. (Newbold, p. 217) Let the random variable $X$ follow a normal distribution with $\mu = 80$ and $\sigma^2 = 100$.

    (a) Find the probability that $X$ is greater than 60.
    (b) Find the probability that $X$ is greater than 72 and less than 82.
    (c) Find the probability that $X$ is less than 55.
    (d) The probability is 0.1 that $X$ is greater than what number?
    (e) The probability is 0.6826 that $X$ is in the symmetric interval about the mean between which two numbers?

20. (Newbold, p. 219) A company services copiers. A review of its records shows that the time taken for a service call can be represented by a normal random variable with a mean of 75 minutes and a standard deviation of 20 minutes.

    (a) What proportion of service calls takes less than 1 hour?
    (b) What proportion of service calls takes more than 90 minutes?
    (c) Sketch a graph to show why the answers to parts (a) and (b) are the same.

(d) The probability is 0.1 that a service call takes more than how many minutes?

21. Do Portuguese people sleep enough hours? According to a study, the number of hours slept on a typical night by Portuguese people is normally distributed with an average of 6.8 hours and a standard deviation of 0.6 hours.

    (a) What is the probability that a randomly selected person sleeps more than 8 hours?
    (b) What is the probability that a randomly selected person sleeps 6 hours or less?
    (c) Doctors suggest sleeping between 7 and 9 hours. What percentage of the population gets this much sleep?
    (d) For you to be classified in the top 10% of Portuguese people that sleep the most, how many hours would you have to sleep on a typical night?

22. In the course of an investigation regarding a complaint about the warranty period of the ROADSTER brand of automobile tires, a consumer safety commission is analyzing the quality of the production process. The manufacturer gives a 45 thousand km warranty on these tires. However, it also knows that due to variability in the production process and tire usage, the duration of a tire under regular usage conditions is random with a normal distribution with mean 50 thousand km and standard deviation 5 thousand km.

    (a) What is the probability that a random tire lasts less than 50 thousand km (you don't need to check the Normal distribution table to answer this)?
    (b) What is the probability that a randomly selected tire will last less than the 45 thousand km given by the warranty?
    (c) What is the probability that a random tire lasts between 49 and 51 thousand km?
    (d) There is a 5% probability that the duration of a tire is "extremely low". What does "extremely low" mean?

23. Consider an investment opportunity such that in four years from today its actual return is described as a normally distributed random variable with an expected value of 30% and a standard deviation of 10%.

    (a) What is the probability that the return will be negative?
    (b) A "pessimistic return" was calculated according to the following criteria: there is a 95% probability that the actual return is above the "pessimistic return". What is the "pessimistic return"?

24. The Breese Company operates a paid highway connecting two major cities. For next year, the price to be paid by automobiles as tolls is already set at 0.10 euros per km. However, there is considerable uncertainty about total automobile circulation. According to several studies that were performed, total circulation (measured in million km) during the next year can be described as a random variable that follows a normal distribution with mean 300 and standard deviation equal to 40. In addition, next year, total costs (in millions of euros) are also random with a normal distribution with mean 26 and standard deviation 4. Assume these two random variables have a joint normal distribution. Total profit is given by: Profit = Price $\times$ Circulation - Costs. It is also known that the correlation between costs and circulation during next year is 0.5. Answer the following questions, explaining and presenting all the necessary calculations.

    (a) What is the probability that the total cost next year is less than 25 million euros?
    (b) What is the expected value of profit next year?
    (c) What is the variance of profit next year?

(d) The probability that next year's profit exceeds a certain value is 5%. What is this value?

25. Mr. Smith is the manager of an octopus fishing company. He has negotiated with a buyer a fixed price of 5000 euros per ton for all fishing in the next month. Although the price is fixed, there is some uncertainty about the total amount that will be caught and also on total costs. Based on his knowledge of fishing, Mr. Smith concluded that the total amount (in tons) that will be caught next month can be described as a random variable that follows a normal distribution with mean 20 and standard deviation equal to 2. Mr. Smith also believes that next month the total costs (in thousands of euros) are random with a normal distribution with mean 50 and standard deviation 10. The total amount of octopus that is caught and the total costs are correlated with a correlation coefficient equal to 0.1. Assume the two random variables have a joint normal distribution. Based on this information, we intend to study the randomness of the total profits for the next month (profit = price $\times$ quantity - cost). Answer the following questions.

    (a) What is the probability that total costs in the next month exceed 60 thousand euros?
    (b) What is the expected value of total profits in the coming month?
    (c) What is the variance of total profits in the coming month?
    (d) For what value of the correlation between total quantity and total costs is the variance of total profits maximum?

26. You have 100 euros to invest in two alternative options denoted by A and B. Let $Q_A$ and $Q_B$ denote the amount of euros invested in A and B, respectively. You must have $Q_A + Q_B = 100$, $Q_A \geq 0$, and $Q_B \geq 0$. Let $X_A$ and $X_B$ denote the profits obtained for each euro invested in A and B, respectively. Therefore, total profit is given by $Q_A X_A + Q_B X_B$. Suppose that $X_A$ and $X_B$ are independent random variables with a normal distribution and have the same expected value given by 3 euros. However, the standard deviation of $X_A$ equals 3, while the standard deviation of $X_B$ equals 2.

    (a) What is the probability that $X_A$ is positive?
    (b) What is the probability that $X_A$ is greater than $X_B$?
    (c) Which of the following two investment strategies is preferred? Invest everything in alternative B? Or a 50-50 split between alternatives A and B? Justify your answer.
    (d) Assume the two random variables have a joint normal distribution. If $X_A$ and $X_B$ were negatively correlated, how would your answer to question (c) change?

27. Rita Summers, executive manager of Creative Creations, Inc., operates a design company that sells its creative designs to advertising companies in Euroland. Rita is planning the development of a new branch of the company in Southland. She thinks that it will be possible to charge a fixed price of 2 thousand euros for each new design sold in this market. However, there is some uncertainty regarding the actual market demand. Rita believes that the number of designs that can be sold per month is normally distributed with a mean of 100 and a standard deviation of 5. Monthly revenues are given by the number of designs sold multiplied by the unit price of 2 thousand euros. Monthly profits are given by monthly revenues minus monthly costs.

    (a) What is the probability that in a given month, it is possible to sell at least 99 designs in Southland?
    (b) Suppose that monthly costs for the Southland branch will be fixed and equal to 190 thousand euros. Determine:
        i. The expected monthly profits in $10^3$ euros.
        ii. The standard deviation of monthly profits in $10^3$ euros.
        iii. The probability distribution of monthly profits in $10^3$ euros.

iv. The probability that in a given month profits are negative.

(c) Suppose now that both monthly sales and costs are uncertain and have a joint normal distribution. In particular, Rita thinks costs have a mean of 190 thousand euros and a standard deviation of 10 thousand euros. Since the success of the new branch will depend on the quality of the creative designers hired to work at the Southland branch, it is likely that costs (in particular, personnel costs) and sales are correlated. Rita is not yet sure about the value of this correlation - assume it is equal to $\rho$. Determine:

i. The expected monthly profits in $10^3$ euros.

ii. The standard deviation of monthly profits in $10^3$ euros.

iii. The probability distribution of monthly profits in $10^3$ euros.

iv. The distribution of monthly profits if costs and sales are perfectly and positively correlated.

28. Suppose that you are responsible for managing the sales department of a computer company with two shops in Lisbon. In the following month, these two stores will put on sale the new UniPad tablet. You must decide today how many units should be purchased from the manufacturer in order to jointly meet the demand in the two shops in Lisbon in July. The demands of UniPad tablets in the two stores in the following month are uncertain and may be considered as random variables. The demand in the first store can be described by a normal distribution with mean 40 and standard deviation 4. The demand in the second store can be described by a normal distribution with mean 60 and standard deviation 3. Assume these two random variables have a joint normal distribution.

(a) What is the probability that the demand for UniPad tablets in the first store exceeds 46 units next month?

(b) If the correlation between demands in the two stores equals 0.5, what is the variance of the total demand for the UniPad tablets in the two stores (given by the sum of the demands in the first and second stores) next month?

(c) If the total demand in the two stores is below the number of units purchased from the manufacturer, there will be a loss. On the other hand, if the total demand exceeds the number of units purchased to the manufacture then it will not be possible to satisfy all the demand and some of your clients will be unhappy. Answer the following two questions:

i. Suppose it is possible to assume that in July the demands for UniPad tablets in the two stores are independent of each other. How many units should you purchase from the manufacturer so that you can satisfy total demand (the sum of the demands in the first and second stores) with a probability of 85%?

ii. If the demands of UniPads in the two stores are correlated, what is the value of the correlation which would require a maximum purchase from the manufacturer to ensure it is possible to meet total demand with a probability of 85%?

## 2 Sampling distributions

1. Prove that if $X_i, i = 1, 2, ..., n$ are i.i.d., with p.d.f. $f(x)$, $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$, then:

   (a) $E(\bar{X}) = \mu$

   (b) $Var(\bar{X}) = \sigma^2/n$

2. Let $p$ be the population proportion and $\hat{p}$ the sample proportion. Prove that:

   (a) $E(\hat{p}) = p$

   (b) $Var(\hat{p}) = \frac{p(1-p)}{n}$

3. If the population mean $\mu$ was known, then it would be possible to estimate the population variance $\sigma^2$ using an alternative formula for the sample variance: $\hat{\sigma}^2 = \sum_{i=1}^{n}(X_i - \mu)^2/n$. Prove that $E(\hat{\sigma}^2) = \sigma^2$.

4. A fast food restaurant chain operating in shopping malls in all major cities in Portugal, wants to know the opinion of shopping mall visitors about the quality of their food. One option consists in asking their clients to fill out a simple questionnaire before they leave the restaurant. Discuss the reasons why this may not result in a simple random sample and what can go wrong when you try to interpret the survey results.

5. A tourist agency wants to know what proportion of visitors to the Mosteiro dos Jerónimos are from Portuguese speaking countries. To find out, they survey 20 people in the line to purchase tickets one Sunday afternoon in July. Will that result in a simple random sample? Are all the assumptions and conditions for finding the sampling distribution of the proportion satisfied? Explain.

6. As responsible for the human resources management at a major real estate company, you are considering offering all your sales agents a one-week workshop on negotiation techniques. However, you are unsure of how much will your agents really benefit from attending that workshop. Therefore, you decide to start by selecting only a sample of your employees to go to that workshop. You send out an email invitation to all your agents stating that only the first 20 applications will be considered. Then, two months after these 20 agents have taken the workshop, you measure by how much has their performance increased. Will this be a simple random sample from the population of all sales agents in your company? What kind of problems may arise with this sampling approach? How could you select a simple random sample?

7. Market analysts often use data on purchases with supermarket discount cards to understand consumer habits and preferences. A known network of hypermarkets operating in Portugal wants to study the demand for games and toys. As the database of all its customers holding the card is too big, it was decided to use a random sample of 1600 of these customers. One of the findings was that in the last week, 1 in every 10 clients in this sample bought some game or toy. Consider the following question: "What can you conclude about the spending habits of all Portuguese consumers regarding purchases of games and toys?". Is this a valid sample to answer this question? Indicate one strong point and two weak points of this sample.

8. Consider a random variable $X$ whose distribution in the population is Normal with mean $\mu = 50$ and standard deviation $\sigma = 10$. You are going to select a simple random sample of size $n = 30$, denoted by $(X_1, X_2, ..., X_{30})$, from that population and compute its sample mean, $\bar{X}$.

   (a) What is the probability that the first observation, $X_1$, will be between 49 and 51?

   (b) What is the probability that the second observation, $X_2$, will be between 49 and 51?

   (c) What is the expected value of $\bar{X}$?

(d) What is the standard deviation of $\bar{X}$?

(e) What is the probability that the sample mean, $\bar{X}$, will be between 49 and 51?

(f) There is a 2.5% chance that $\bar{X}$ will be above some value. What is that value?

(g) Suppose you already know the value of the first observation. Why is your answer to question (b) unaffected?

(h) If the distribution of the random variable $X$ in the population was not normal, even though with the same mean and variance as mentioned above, how would that affect your answers to each of the above questions?

9. Maria Albright is the Human Resources director of a large accounting firm. Every month, she has to evaluate the performance of several randomly selected employees. From past experience, Maria believes that the time it takes to perform the evaluation of each employee is a random variable with a normal distribution with a mean of 5 hours and a standard deviation of 1.5 hours.

   (a) What is the probability that an evaluation takes between 4.5 and 5.5 hours?

   (b) What is the probability that 9 evaluations take an average time between 4.5 and 5.5 hours?

   (c) What is the probability that 9 evaluations take a total time between 40 and 50 hours?

   (d) If instead of 9 evaluations, it is necessary to carry out 20, how do the probabilities calculated in questions (b) and (c) change? Indicate only if they remain the same, decrease or increase, without further calculations.

   (e) The last 9 evaluations took an average time of 5.5 hours to complete. Is this result plausible given Maria's hypothesis of an average time of completion of 5 hours? What can you conclude?

   (f) The last 100 evaluations took an average time of 5.5 hours to complete. Is this result plausible given Maria's hypothesis of an average time of completion of 5 hours? What can you conclude?

10. A manufacturer of LED lamps advertises an average lifetime of 50 thousand hours. The company also acknowledges that the lifetime of any particular LED lamp may vary due to some variability in the production process and in the usage conditions. According to an engineering study, the variability in lifetime can be described by a normal distribution with a standard deviation equal to 12 thousand hours. An inspection team is going to randomly select 9 LED lamps and test their lifetime. If they find a sample mean below 45 thousand hours they will have to stop the production process and to conduct a complete verification of all the manufacturing equipment which will of course involve large costs. The company truly beliefs in the advertised average lifetime and in the engineering study report. Should the company be worried about the result of the inspection?

    (a) Provide an answer to that question using your knowledge of the distribution of the sample mean.

    (b) Instead of relying on the statistical distribution of the sample mean, you decide to analyze this question using the following simulation approach. Use Excel to generate many random samples of size $n = 9$ (what is meant by many? Ideally $n$ should be infinity! But you can generate 1000 random samples). For each random sample compute the corresponding sample mean. Analyze the distribution of the simulated sample means by plotting an histogram, and computing the mean and standard deviation. Also compute the proportion of sample means that are below the 45 thousand threshold. Are these results in accordance with the theoretical (and correct) results obtained in question (a) above?

11. The Gini index provides a measure of the degree of inequality in the distribution of family income in a country. It measures inequality on a scale from 0 to 100, where higher values indicate higher

inequality. If income were distributed with perfect equality, the index would be zero; if income were distributed with perfect inequality, the index would be 100. You can find some data on this in the file *Gini.xlsx*.

(a) Plot the histogram of the Gini index for all the countries (this is the population distribution) and compute the mean, variance, standard deviation (these are some parameters describing that population distribution). Is the distribution close to a Normal distribution? Is it symmetrical?

(b) If you were to randomly select 5 countries (a sample) from the list of countries you downloaded (the population), and computed the mean of the Gini indices for those countries (the sample mean), what would be the expected value of that sample mean? And the variance of that sample mean? What about the probability distribution?

(c) Select one random sample of 5 countries and compute the mean of the Gini indices for those countries. If you select another random sample do you obtain the same value? Why?

12. A tire manufacturing company gives a warranty of 95,000 kilometers in any of their SHAPER brand tires. However, the company knows that due to variability in the production process and tire usage, the duration of any specific tire is random with a normal distribution, an expected value equal to 100,000 kilometers, and a standard deviation of 5,000 km.

(a) What is the probability that a randomly selected tire lasts less than 95,000 km?

(b) The probability that a randomly selected tire has an "extremely high" duration is 5%. What is meant by "extremely high"?

(c) Suppose four tires are going to be randomly and independently chosen. The average duration of these four tires will be calculated.

i. What is the expected average duration of the four tires?

ii. What is the standard deviation of the average duration of the four tires?

iii. How likely is the average duration of the four tires to be over 95,000 km?

(d) Suppose now that the four tires will be all chosen from the same car. In this situation, the correlations between the durations of any pair of tires will be positive. How do the answers to (c) i., ii., and iii. change?

13. Nielsen wants to estimate the average number of hours Americans spend surfing the web per month. This average is the same in the men's and in the women's populations. However, the variance in the men's population is 3 times larger than in the women's population. Calculate the variance of the sample mean for a random sample of 200 Americans, composed of 100 men and of 100 women, where the men and women samples were selected independently of each other.

14. Referring to exercise 13, denote by $\bar{X}_M$ the sample mean for the 100 men and by $\bar{X}_W$ the sample mean for the 100 women. Consider an estimator of the overall population mean given by

$$a\bar{X}_M + (1-a)\bar{X}_W$$

Which value of $a \in [0,1]$ appearing in the previous formula minimizes the variance of the estimator?

15. (Newbold, p. 262) Given a population with a mean of $\mu = 100$ and a variance of $\sigma^2 = 81$, the central limit theorem applies when the sample size is $n \geq 25$. A random sample of size $n = 25$ is obtained.

(a) What are the mean and variance of the sampling distribution for the sample means?

(b) What is the probability that $\bar{X} > 102$?

(c) What is the probability that $98 \leq \bar{X} \leq 101$?

(d) What is the probability that $\bar{X} \leq 101.5$?

16. (Newbold, p. 262) Given a population with a mean of $\mu = 100$ and a variance of $\sigma^2 = 900$, the central limit theorem applies when the sample size is $n \geq 25$. A random sample of size $n = 30$ is obtained.

    (a) What are the mean and variance of the sampling distribution for the sample means?

    (b) What is the probability that $\bar{X} > 109$?

    (c) What is the probability that $96 \leq \bar{X} \leq 110$?

    (d) What is the probability that $\bar{X} \leq 107$?

17. (Newbold, p. 263) The fuel consumption, in miles per gallon, of all cars of a particular model has a mean of 25 and a standard deviation of 2. The population distribution can be assumed to be normal. A random sample of these cars is taken.

    (a) Find the probability that sample mean fuel consumption will be fewer than 24 miles per gallon if

        i. a sample of 1 observation is taken.

        ii. a sample of 4 observations is taken.

        iii. a sample of 16 observations is taken.

    (b) Explain why the three answers in part (a) differ in the way they do. Draw a graph to illustrate your reasoning.

18. (Newbold, p. 263) The mean selling price of senior condominiums in Green Valley over a year was \$215,000. The population standard deviation was \$25,000. A random sample of 100 new unit sales was obtained.

    (a) What is the probability that the sample mean selling price was more than \$210,000?

    (b) What is the probability that the sample mean selling price was between \$213,000 and \$217,000?

    (c) What is the probability that the sample mean selling price was between \$214,000 and \$216,000?

    (d) Without doing the calculations, state in which of the following ranges the sample mean selling price is most likely to lie: \$213,000 to \$215,000; \$214,000 to \$216,000; \$215,000 to \$217,000; \$216,000 to \$218,000

    (e) Suppose that, after you had done these calculations, a friend asserted that the population distribution of selling prices of senior condominiums in Green Valley was almost certainly not normal. How would you respond?

19. (Newbold, p. 263) Candidates for employment at a city fire department are required to take a written aptitude test. Scores on this test are normally distributed with a mean of 280 and a standard deviation of 60. A random sample of nine test scores was taken.

    (a) What is the standard error of the sample mean score?

    (b) What is the probability that the sample mean score is less than 270?

    (c) What is the probability that the sample mean score is more than 250?

    (d) Suppose that the population standard deviation is, in fact, 40, rather than 60. Without doing the calculations, state how this would change your answers to parts (a), (b), and (c). Illustrate your conclusions with the appropriate graphs.

20. (Newbold, p. 264) The times spent studying by students in the week before final exams follows a normal distribution with standard deviation 8 hours. A random sample of four students was taken in order to estimate the mean study time for the population of all students.

    (a) What is the probability that the sample mean exceeds the population mean by more than 2 hours?

    (b) What is the probability that the sample mean is more than 3 hours below the population mean?

    (c) What is the probability that the sample mean differs from the population mean by more than 4 hours?

    (d) Suppose that a second (independent) random sample of 10 students was taken. Without doing the calculations, state whether the probabilities in parts (a), (b), and (c) would be higher, lower, or the same for the second sample.

21. (Newbold, p. 264) An industrial process produces batches of a chemical whose impurity levels follow a normal distribution with standard deviation 1.6 grams per 100 grams of chemical. A random sample of 100 batches is selected in order to estimate the population mean impurity level.

    (a) The probability is 0.05 that the sample mean impurity level exceeds the population mean by how much?

    (b) The probability is 0.10 that the sample mean impurity level is below the population mean by how much?

    (c) The probability is 0.15 that the sample mean impurity level differs from the population mean by at least how much?

22. (Newbold, p. 264) The number of hours spent studying by students on a large campus in the week before final exams follows a normal distribution with a standard deviation or 8.4 hours. A random sample of these students is taken to estimate the population mean of the number of hours studying.

    (a) How large a sample is needed to ensure that the probability that the sample mean differs from the population mean by more than 2.0 hours is less than 0.05?

    (b) Without doing the calculations, state whether a larger or smaller sample size compared to the sample size in part (a) would be required to guarantee that the probability of the sample mean differing from the population mean by more than 2.0 hours is less than 0.10.

    (c) Without doing the calculations, state whether a larger or smaller sample size compared to the sample size in part (a) would be required to guarantee that the probability of the sample mean differing from the population mean by more than 1.5 hours is less than 0.05.

23. Consider an unbalanced coin that has a probability of giving heads equal to 2/3. You are going to flip that coin several times and calculate the proportion of heads. You can assume that the results of the coin flips are independent.

    (a) What is the probability that in 3 flips of the coin, the proportion of heads is less than or equal to 0.5?

    (b) What is the probability that in 100 flips of the coin, the proportion of heads is less than or equal to 0.5?

24. The chain of restaurants H2O2 regularly conducts studies of customer satisfaction. Based on these studies, the marketing department of the company publicly advertises that 90% of all its customers assigned a grade of "excellent" to the quality of the food being served. A consumer

rights protection magazine will analyze this statement by performing a survey of 100 randomly chosen customers. Assuming that the company's claim about the population of their customers is indeed correct, answer the following questions.

(a) What is the probability that less than 80 out of the 100 customers in the sample give a grade of "excellent"?

(b) What should be the sample size to guarantee that the probability calculated in the previous paragraph is below 5%?

25. Study the properties of the sample mean using Monte Carlo simulation and statistical theory. Consider the following experiments (always assume that any sample of $n$ observations: $X_1$, $X_2$,..., $X_n$ is i.i.d.):

- The "good" coin experiment: $X = 0$ or 1 with $P(X = 1) = 0.5$;
- The "bad" coin experiment: $X = 0$ or 1 with $P(X = 1) = 0.2$;
- The "good" dice experiment: $X = 1, 2, 3, 4, 5, 6$ with $P(X = i) = 1/6$;
- The "bad" dice experiment: $X = 1, 2, 3, 4, 5, 6$ with $P(X = 6) = 0.5$ and $P(X = i) = 0.1, i = 1, 2, 3, 4, 5$;
- Stock exchange simulation: Assume that the percent change in the stock index in one day equals $X \cap N(\mu, \sigma^2)$ with $\mu = 1$ and $\sigma^2 = 2^2$.

(a) For each of the above cases, complete the following tasks in Excel:

i. Generate 1000 samples where each sample is i.i.d. with $n$ observations generated from one of the above experiments.

ii. Compute the average of the $n$ observations for each of the 1000 samples (also called the sample mean).

iii. Produce an histogram of the 1000 sample means.

iv. Calculate the average of the 1000 sample means.

v. Calculate the variance of the 1000 sample means.

(b) Answer the following tasks using your knowledge of Statistics:

i. What is the distribution of probabilities of one single observation?

ii. What can you say about the distribution of probabilities of the sample mean? Compare it with the result obtained in (a)iii.

iii. What is the expected value of the sample mean? Compare it with the number obtained in (a)iv.

iv. What is the variance of the sample mean? Compare it with the number obtained in the answer to (a)v.

26. An opinion poll regarding the attitudes of Portuguese drivers will be conducted soon. A random sample of 100 drivers has already been selected for this poll. One of the questions included in the survey is the following: "Do you think that you drive better than the median Portuguese driver?". Each respondent will answer with "Yes" or "No". If exactly half of the Portuguese drivers' population thinks that they drive better than the median driver, what is the probability that more than 55 drivers in the sample think that they drive better than the median driver?

27. (Newbold, p. 269) A record store owner finds that 20% of customers entering her store make a purchase. One morning 180 people, who can be regarded as a random sample of all customers, enter the store.

(a) What is the mean of the distribution of the sample proportion of customers making a purchase?

(b) What is the variance of the sample proportion?

(c) What is the standard error of the sample proportion?

(d) What is the probability that the sample proportion is less than 0.15?

28. (Newbold, p. 269) A charity has found that 42% of all donors from last year will donate again this year. A random sample of 300 donors from last year was taken.

    (a) What is the standard error of the sample proportion who will donate again this year?

    (b) What is the probability that more than half of these sample members will donate again this year?

    (c) What is the probability that the sample proportion is between 0.40 and 0.45?

    (d) Without doing the calculations, state in which of the following ranges the sample proportion is more likely to lie: 0.39 to 0.41, 0.41 to 0.43, 0.43 to 0.45, or 0.45 to 0.46.

29. (Newbold, p. 278) A sample of 100 students is to be taken to determine which of two brands of beer is preferred in a blind taste test. Suppose that, in the whole population of students, 50% would prefer brand A.

    (a) What is the probability that more than 60% of the sample members prefer brand A?

    (b) What is the probability that between 45% and 55% of the sample members prefer brand A?

    (c) Suppose that a sample of only 10 students was available. Indicate how the method of calculation of probabilities would differ, compared with your solutions to parts (a) and (b)?

30. (Newbold, p. 269) A random sample of 100 voters is taken to estimate the proportion of a state's electorate in favor of increasing the gasoline tax to provide additional revenue for highway repairs. What is the largest value that the standard error of the sample proportion in favor of this measure can take?

31. (Newbold, p. 270) In the previous exercise, suppose that it is decided that a sample of 100 voters is too small to provide a sufficiently reliable estimate of the population proportion. It is required instead that the probability that the sample proportion differs from the population proportion (whatever its value) by more than 0.03 should not exceed 0.05. How large a sample is needed to guarantee that this requirement is met?

32. (Newbold, p. 270) A company wants to estimate the proportion of people who are likely to purchase electric shavers from those who watch the nationally telecast baseball playoffs. A random sample obtained information from 120 people who were identified as persons who watch baseball telecasts. Suppose that the proportion of those likely to purchase electric shavers in the population who watch the telecast is 0.25.

    (a) The probability is 0.10 that the sample proportion watching the telecast exceeds the population proportion by how much?

    (b) The probability is 0.05 that the sample proportion is lower than the population proportion by how much?

    (c) The probability is 0.30 that the sample proportion differs from the population proportion by at least how much?

33. (Newbold, p. 270) The anual percentage salary increases for the chief executive officers of all midsize corporations are normally distributed with mean 12.2% and standard deviation 3.6%. A random sample of 81 of these chief executive officers was taken. What is the probability that more than half the sample members had salary increases of less than 10%?

34. Solve the "Furniture fire case study".

# 3 Point estimation

1. (Newbold, p. 290) A random sample of 10 economists produced the following forecasts for percentage growth in real gross domestic product in the next year:

$$2.2 \ 2.8 \ 3.0 \ 2.5 \ 2.4 \ 2.6 \ 2.5 \ 2.4 \ 2.7 \ 2.6$$

   Use unbiased estimation procedures to find point estimates for the following:

   (a) The population mean
   (b) The population variance
   (c) The variance of the sample mean
   (d) The population proportion of economists predicting growth of at least 2.5% in real gross domestic product

2. (Newbold, p. 290) Suppose that $X_1$ and $X_2$, are random samples of observations from a population with mean $\mu$ and variance $\sigma^2$. Consider the following three point estimators $\hat{\mu}_1$, $\hat{\mu}_2$ and $\hat{\mu}_3$ of $\mu$:

$$\hat{\mu}_1 = \frac{X_1 + X_2}{2}; \ \hat{\mu}_2 = \frac{X_1 + 3X_2}{4}; \ \hat{\mu}_3 = \frac{X_1 + 2X_2}{3}$$

   (a) Show that all three estimators are unbiased.
   (b) Which of the estimators is the most efficient?
   (c) Find the relative efficiency of $\hat{\mu}_1$ with respect to each of the other two estimators.

3. Suppose that $X_1$, $X_2$ consists of a simple random sample of size $n = 2$, obtained from a population with mean $\mu$ and variance $\sigma^2$. Consider the following possible estimators of $\mu$:

$$\hat{\mu} = \bar{X}; \tilde{\mu} = \frac{X_1}{4} + \frac{3X_2}{4}; \bar{\mu} = \frac{X_1}{3} + \frac{X_2}{3}$$

   (a) Which of the following statements is correct?
      i. Only $\hat{\mu}$ is unbiased
      ii. Only $\tilde{\mu}$ is unbiased
      iii. All are unbiased
      iv. None of the previous statements is correct
   (b) Which of the following statements is correct?
      i. The variances of $\hat{\mu}$ and $\tilde{\mu}$ are equal
      ii. The variance of $\tilde{\mu}$ is less than the variance of $\hat{\mu}$
      iii. The variance of $\hat{\mu}$ is less than the variance of $\tilde{\mu}$
      iv. The difference between the variances of $\hat{\mu}$ and $\tilde{\mu}$ can be positive, zero, or negative, depending on the particular values taken by the mean $\mu$ and variance $\sigma^2$

4. From some population with Normal distribution, we will obtain a sample with just two observations. The mean of this population is unknown. The variance is known and equals 12. One wants to estimate the mean of the population from that sample. Although the two observations have the same distribution, they are not independent. It is known that they have a joint normal distribution and their correlation equals 0.5. Characterize the sample mean in terms of its:

   (a) Expected Value
   (b) Variance
   (c) Distribution

5. Two independent studies are going to be undertaken with the objective of estimating the value of the black market economy in Portugal during 2012. Both studies will use unbiased estimators to estimate that value. It is known that the first study will result in an estimator, denoted by $U$, with half the standard deviation of the estimator from the second study, denoted by $V$. You will be able to access the results of both studies. You want to use both estimators to construct a new and more efficient estimator using the following formula: $W = aU + bV + c$, where $a, b$, and $c$ are some constants.

   (a) Assuming that both $U$ and $V$ are uncorrelated, answer the following questions. Present and justify all your derivations.

      i. What values of the constants $a$, $b$ and $c$ guarantee that $W$ is unbiased?

      ii. What values of the constants $a$, $b$ and $c$ guarantee that $W$ is unbiased and has minimal variance?

   (b) Assume now that the estimators coming from the two studies are correlated. Suppose the correlation is equal to -1. What values of the constants $a$, $b$ and $c$ guarantee that $W$ is unbiased and has minimal variance? Present and justify all your derivations.

6. We want to study a random variable $X$ with a normal distribution in the population. For this purpose, a sample with four observations will be used. The observations in this sample are independent and all have the same expected value. Suppose the variance of each observation in the sample is increasing. Specifically, the variance of the 1st observation equals a certain value of $\sigma^2$ the variance of the 2nd observation is equal to twice the variance of the 1st observation, the variance of the 3rd observation is three times the variance of the 1st observation, and the variance of the 4th observation is four times the variance of the 1st observation.

   Suppose we know that the expected value of $X$ is equal to 50. Propose an unbiased estimator of $\sigma^2$ using all observations in the sample, but with different weights for each observation. Demonstrate that the proposed estimator is, in fact, unbiased.

7. An American chain of restaurants will start its business in Europe by launching a number of restaurants in major European cities. The profits for each restaurant will depend on market conditions, demand, and competition in each city. The profit of a restaurant can be considered as a random variable with expected value equal to 100 and standard deviation equal to 1000. The profits at the various restaurants are mutually independent. Consider the following three alternatives in terms of the number of restaurants to launch, denoted by $n$:

   - $n = 1$ restaurant
   - $n = 100$ restaurants
   - $n = 1000$ restaurants

   These three alternatives need to be evaluated in terms of the (simple arithmetic) average profit across the $n$ restaurants.

   (a) Which of the three alternatives leads to a higher expected value of the average profit?

   (b) Which of the alternatives leads to a lower variance of the average profit?

8. The designers working at an advertising agency proposed a new package for a well-known orange juice. The current package will be denoted as $A$, and the new package by $B$. To analyze consumer preferences for these two packages, a trial will be conducted where each participating consumer will provide a rating between 0 and 100 to each package. The objective is to estimate the average difference in ratings given to the two packages for the whole population.

   There are two ways to implement this trial. The first way is to collect independently two random samples of $n$ consumers each (requiring a total of $2n$ consumers) and ask consumers in one of the

samples to rate the $A$ package, and ask consumers in the another sample to rate the $B$ package. The obvious estimator of the average difference in the population is the difference between the means in the two samples.

Another way to implement the trial is to collect a random sample of $n$ consumers and ask each of the consumers to rate both packages: A and B. The estimator in this case is the mean of the $n$ differences in ratings, of $A$ vs. $B$, in the sample of $n$ consumers. It is plausible that for the same consumer, the ratings given to the two packages ($A$ and $B$) are not mutually independent. This is because individuals with a tendency to provide higher ratings should give higher ratings to both $A$ and $B$, and those that in general provide lower ratings should give lower ratings to both $A$ and $B$.

Which of the two estimators described above is preferable in terms of:

(a) Expected value.

(b) Variance.

# 4 Confidence intervals

1. (Newbold, p. 296) Assume a normal distribution with known population variance. Calculate the lower confidence limit (LCL) and upper confidence limit (UCL) for a confidence interval for $\mu$ for each of the following.

   (a) $\bar{x} = 50$; $n = 64$; $\sigma = 40$; $\alpha = 0.05$
   (b) $\bar{x} = 85$; $n = 225$; $\sigma^2 = 400$; $\alpha = 0.01$
   (c) $\bar{x} = 510$; $n = 485$; $\sigma = 50$; $\alpha = 0.10$

2. (Newbold,p. 302) Assuming a normal distribution, find the LCL and UCL for a confidence interval for $\mu$ for each of the following:

   (a) $\alpha = 0.05$; $n = 25$; $\bar{x} = 560$; $s = 45$
   (b) $\alpha/2 = 0.05$; $n = 9$; $\bar{x} = 160$; $s^2 = 36$
   (c) $1 - \alpha = 0.98$; $n = 22$; $\bar{x} = 58$; $s = 15$

3. Find the margin of error of the confidence interval for the mean of a normal population for each of the following:

   (a) $n = 20$; 90% confidence level; $s = 36$
   (b) $n = 7$; 98% confidence level; $s = 16$
   (c) $n = 16$; 95% confidence level; $s^2 = 43$
   (d) 99% confidence level; $x_1 = 15; x_2 = 17; x_3 = 13; x_4 = 11$.

4. (Newbold, p. 296) A personnel manager has found that historically the scores on aptitude tests given to applicants for entry-level positions follow a normal distribution with a standard deviation of 32.4 points. A random sample of nine test scores from the current group of applicants had a mean score of 187.9 points.

   (a) Find an 80% confidence interval for the population mean score of the current group of applicants.
   (b) Based on these sample results, a statistician found for the population mean a confidence interval extending from 165.8 to 210.0 points. Find the confidence level of this interval.

5. To study the number of hours that college students sleep on average, we selected a random sample of 25 students. Below, are presented some descriptive statistics regarding the number of hours each student slept the night before:

   - Minimum = 5.8
   - Maximum = 9.5
   - Average = 7.0
   - Median = 6.5
   - Standard Deviation = 1.0

   Answer the following questions assuming that the population distribution is normal.

   (a) Suppose we knew that the standard deviation of hours slept by all students in the population was equal to 0.5. What is the 95% confidence interval for the average hours slept by all students in the population? If you want the margin of error of the 95% confidence interval to be 0.1 hours, what should be the sample size?

(b) Suppose now that the variance of hours slept by all students in the population was unknown. Using the statistics presented above regarding the sample of 25 students, what is the 95% confidence interval for the average hours slept by all students in the population?

(c) What is the meaning of a 95% confidence interval for the average hours slept by all students in the population?

i. 95% of the students in the population slept a number of hours inside that interval

ii. 95% of the students in the sample slept a number of hours inside that interval

iii. the probability that a random sample results in a sample mean with a value inside that interval is equal to 95%

iv. all previous statements are incorrect

6. A process producing bricks is known to give output whose weights are normally distributed with standard deviation of 0.12 Kg. A random sample of 16 bricks from today's output had a mean weight of 4.07 Kg.

(a) Find a 99% confidence interval for the mean weight of all bricks produced this day.

(b) Without doing the calculations, explain whether a 95% confidence interval for the population mean would be wider than, narrower than, or have the same width as that found in part (a).

(c) It is decided that tomorrow a sample of 20 bricks will be taken. Without doing the calculations, explain whether a correctly calculated 99% confidence interval for the mean weight of tomorrow's output would be wider than, narrower than, or the same width as that found in part (a).

(d) Suppose that the population standard deviation for today's output is 0.15 Kg (not 0.12 Kg). Without doing the calculations, explain whether a correctly calculated 99% confidence interval for the mean weight of today's output would be wider than, narrower than, or the same width as that found in part (a).

7. (Newbold, p. 324) A random sample of 174 college students was asked to indicate the number of hours per week that they surf the Internet for either personal information or material for a class assignment. The sample mean response was 6.06 hours and the sample standard deviation was 1.43 hours. Based on these results, a confidence interval extending from 5.96 to 6.16 was calculated for the population mean. Find the confidence level of this interval.

8. (Newbold, p. 325) A test was taken by 90 students. A random sample of 10 scores found the following results:

93 71 62 75 81 63 87 59 84 72

(a) Find a 90% confidence interval for the population's mean score.

(b) Without doing the calculations, state whether a 95% confidence interval for the population mean would be wider or narrower than the interval found in part (a).

9. (Newbold, p. 323) An automobile dealer has an inventory of 400 used cars. To estimate the mean mileage of this inventory, she intends to take a simple random sample of used cars. Previous studies suggest that the population standard deviation is 10,000 miles. A 90% confidence interval for the population mean must extend 2,000 miles on each side of its sample estimate. How large of a sample size is necessary to satisfy this requirement?

10. A mobile phone manufacturing company wants to estimate the average battery life of a new model. To this end, the company has tested 16 of these new phones. These were randomly chosen from a large batch of manufactured new phones. We present some descriptive statistics concerning the duration (in hours) of the batteries of these 16 phones:

- Min. = 20
- Max. = 30
- Average = 25
- Median = 25
- Standard Deviation = 2

Suppose one can assume that the battery life of a mobile phone chosen at random can be described as a random variable with normal distribution. Answer the following questions justifying all the necessary calculations.

(a) Present a 95% confidence interval for the mean duration of new phones.

(b) Suppose for the initial sample of 16 phones, the following confidence interval for the mean battery was presented: [23.7, 26.3]. What is the corresponding confidence level?

11. A consumer association decided to investigate whether the soda cans sold by various brands are correctly filled. Let $X$ equal the amount of refrigerant in centilitres actually existing in a can of New-Cola brand chosen at random. Suppose $X$ has a normal distribution with mean $\mu$ and standard deviation $\sigma$. With the aim of estimating the value of $\mu$, a random sample of $n = 9$ cans will be selected. Answer the following questions by presenting and justifying all the necessary calculations.

    (a) Suppose the sample mean was $\bar{x} = 38$. Knowing that $\sigma = 2$, propose a 95% confidence interval for $\mu$.

    (b) If the value of $\sigma$ was unknown and had to be estimated, how would the amplitude of the new confidence interval change? Explain.

    (c) One of your colleagues proposes the following confidence interval for $\mu$: $(\bar{X} - 1.1, \bar{X} + 1.1)$, where $\bar{X}$ denotes the sample mean. Suppose $\sigma$ is known and equals 2.

        i. What is the confidence level of the proposed confidence interval?

        ii. To obtain a 95% confidence level using the proposed confidence interval, what should be the value of $n$?

12. One company conducted a medical examination of 25 of its workers chosen at random. Based on the blood pressure exam results, the following 95% confidence interval for mean arterial pressure was obtained: [12.2, 13.8]. Which of the following is a valid interpretation for this confidence interval?

    (a) 95% of the workers in the sample have a blood pressure between 12.2 and 13.8.

    (b) 95% of the workers in the company have a blood pressure between 12.2 and 13.8.

    (c) If the process of selecting a random sample of 25 workers was conducted many times then 95% of the resulting confidence intervals would contain the mean blood pressure of the workers in the company.

    (d) If the random sampling was repeated many times, 95% of the sample means would be between 12.2 and 13.8.

13. According to the latest figures released by the Health Ministry, the average waiting time of patients in the entry list for surgery in public hospitals is 3.5 months. In a study on the functioning of the largest public hospital, based on a random sample of 200 patients, it was found that the average waiting time in that hospital is only 3 months, with a 95% confidence interval of 2.7 to 3.3 months, and a standard deviation of 2.5 months. A reader of this study concluded that 95% of the patients of that hospital have a waiting time below the national average. What's wrong with that conclusion?

14. A recent study from Nielsen states that the average American spends 35 hours a month surfing on the internet. Given that this value was obtained from a random sample with 9 observations and the standard deviation in this sample was equal to 6, construct a 98% confidence interval for the population mean. Assume that the population follows a normal distribution. Justify your answer. Interpret the result.

15. A clothing store is interested in understanding how much college students spend on clothing during the first month of the school year. For a random sample of nine students the mean expenditure was \$157.82, and the sample standard deviation was \$37.89. Assuming that the population is normally distributed, find the margin of error of a 95% confidence interval for the population mean.

16. From some population with a Normal distribution, a sample with only two observations will be obtained. The mean of the population is unknown. The variance is known to be 12. One wants to estimate the mean of that population using that sample. Although the two observations in the sample have the same distribution, they are not independent. Their correlation equals 0.5. Assume observations are jointly normally distributed.

    (a) Obtain the expected value, the variance, and the distribution of the sample mean.

    (b) If the two observed values in the sample are equal to 18 and 22, calculate the resulting confidence interval for the mean of the population using a confidence level of 95%.

17. The usual formula for the confidence interval of the population mean when the population distribution is normal and the variance is known, which is given by $\bar{x} \pm z_{\alpha/2}\sigma/\sqrt{n}$, is valid provided the sample is random with identically and independently distributed observations with a normal distribution. Suppose in reality, the observations were not independent. In particular, suppose all pairwise correlations were positive. Answer the following questions. Assume observations are jointly normally distributed.

    (a) If the usual formula for the 95% confidence interval is used, the real confidence level of the interval will be above, below, or equal to 95%?

    (b) One wanted to calculate a correct confidence interval with the available sample. Suppose all pairwise correlations are known to be 0.5. Propose a new formula for the confidence interval that guarantees a real confidence level of 95%.

18. A market research company conducted recently a survey on hygiene and health. The target population was all Portuguese aged 18 and older. One hundred individuals from that population were selected at random.

    (a) Only 20% of the respondents reported having consumed vitamins in the past 12 months. Present a 95% confidence interval for the true proportion in the population considered.

    (b) In this survey were also addressed other issues such as whether people take cough syrups, pills for headaches, etc. For all possible confidence intervals at 95% which could have been computed, indicate the largest possible margin of error.

19. EDI is the electrical utility company in Highland that is responsible for all the distribution of electrical energy to residential consumers in the country. During 2010, the company launched an major energy saving program directed at all consumers in Highland. A sample of 100 customers was selected at random from the population. For each, the company calculated the variation in electricity consumption in 2010 relative to the previous year (in kWh/day). Next, we present some descriptive statistics for this sample:

    - Min. = -2.5
    - Max. = 1.5

- Average = -0.5
- Median = -0.5
- Standard Deviation = 2

Also, of these 100 customers, 80 had a negative variation in their 2010 consumption when compared to the previous year. Answer the following questions.

(a) Present a 95% confidence interval for the proportion of customers in the population of Highland that reduced their electricity consumption in 2010 relative to the previous year.

(b) Using the same sample of 100 customers, someone presented the following confidence interval for the mean change in electricity consumption for the Highland population: [-0.6; -0.4]. What is the corresponding confidence level?

(c) A similar program is going to be implemented in another country. If one want to obtain for that population a 95% confidence interval for the proportion of customers that reduced their consumption with a margin of error not exceeding 1%, what should be the sample size to consider?

20. Market analysts often use data on purchases with supermarket discount cards to understand consumer habits and preferences. A known network of hypermarkets operating in Portugal wants to study the demand for games and toys. As the database of all its customers holding the card is too big, it was decided to use a random sample of 1600 of these customers. One of the findings was that in the last week, 1 in every 10 clients in this sample bought some game or toy. Consider the following question "what can you conclude about the spending habits of all Portuguese consumers in terms of purchases of games and toys?". Present a 95% confidence interval for the proportion in the population using the usual formula. Comment the result.

21. The University of Michigan School of Business publishes the American Customer Satisfaction Index (ACSI) four times a year. Started in 1994 and based on thousands of costumer interviews, customer satisfaction ratings based on a scale from 0 to 100 are gathered for retailers, supermarkets, financial services, parcel-delivery services, airlines, and so forth. "Understaffed stores, clueless sales clerks, automated phone lines that lead you in circles" are a few reasons why the scores for most companies declined between 1995 and 2000. Concerned about this report, the manager of a national retail store in one community surveyed a random sample of 320 customers. The survey indicated that 80 customers thought that customer services in this store was also on the decline. What conclusions can we draw from this data at a 95% confidence level?

22. (Newbold, p. 305) In a random sample of 95 manufacturing firms, 67 indicated that their company attained ISO certification within the last two years. Find a 99% confidence interval for the population proportion of companies that have been certified within the last 2 years.

23. (Newbold, p. 305) Of a random sample of 250 marketing students, 180 rated a case of résumé inflation as unethical. Based on this information a statistician computed a confidence interval extending from 0.68 to 0.76 for the population proportion. What is the confidence level of this interval?

24. (Newbold, p. 306) It is important for airlines to follow the published scheduled departure times of flights. Suppose that one airline that recently sampled the records of 246 flights originating in Orlando found that 10 flights were delayed for severe weather, 4 flights were delayed for maintenance concerns, and all the other flights were on time.

    (a) Estimate the percentage of on-time departures using a 98% confidence level.

    (b) Estimate the percentage of flights delayed for severe weather using a 98% confidence level.

25. (Newbold, p. 319) How large a sample is needed to estimate the population proportion for each of the following?

(a) $ME = 0.03; \alpha = 0.05$

(b) $ME = 0.05; \alpha = 0.05$

(c) Compare and comment on your answers to parts (a) and (b).

26. (Newbold, p. 319) How large a sample is needed to estimate the population proportion for each of the following?

    (a) $ME = 0.05; \alpha = 0.01$

    (b) $ME = 0.05; \alpha = 0.10$

    (c) Compare and comment on your answers to parts (a) and (b).

27. (Newbold, p. 319) The student government association at a university wants to estimate the percentage of the student body that supports a change being considered in the academic calendar of the university for the next academic year. How many students should be surveyed if a 90% confidence interval is desired and the margin of error is to be only 3%?

28. A market research company conducts every month an opinion survey directed at entrepreneurs regarding their future economic expectations. One of the questions is the following: "Do you expect to have a reduction in your company's economic activity during the next month?".

    (a) Suppose that from the data that will be collected from the next sample of entrepreneurs, you will have to build a confidence interval of 95% for the proportion of entrepreneurs who expect to have a reduction in their economic activity. What is the number of observations necessary to ensure that the length of the confidence interval will not exceed 0.05 (i.e. 5 percentage points)?

    (b) The December survey was based on a random sample of 40 entrepreneurs, of which 30 stated that they expected a reduction in their economic activity.

        i. Calculate a 95% confidence interval for the proportion of entrepreneurs in the population that in December expect to have a decrease in economic activity.

        ii. Someone stated that in December 50% of the entrepreneurs in the population expect to have a decrease in economic activity. What can you conclude based on the result from i.? Explain. (Note: a more complete answer to this question requires an hypothesis testing approach that will be covered in the next topic).

        iii. The survey also asked each entrepreneur their forecast about the change in sales compared to the same year last month. Assuming that this variable has a normal distribution in the population, present a 95% confidence interval for its population mean knowing that the sample mean was -10 and the sample variance was 12.

        iv. In the November survey, the sample included 50 entrepreneurs and the result was that 25 of them expected to have a reduction in their economic activity. Suppose this sample was obtained independently from the December sample. Present a 95% confidence interval for the variation between November and December of that proportion in the population of Portuguese entrepreneurs.

29. To evaluate the flavor of a new formula for a well known beer, we set up an experiment with a random sample of 9 consumers. Each individual tasted both the old and the new formula. At the end, each gave a score on the scale of 0 to 20. Results are presented in the table below (consumer names are not real). Construct a 95% confidence interval for the difference between the average scores of the two formula of beer. What hypothesis are necessary in order for the proposed interval to be valid? Provide additional comments regarding the validity of this experiment.

| Consumer name | Old formula score | New formula score |
|---|---|---|
| António Neves | 12 | 14 |
| João Mário | 15 | 15 |
| Maria da Assunção | 15 | 17 |
| Isabel Queiroz | 17 | 18 |
| Óscar da Silva | 15 | 14 |
| Fernando Santos | 13 | 16 |
| Clotilde Magalhães | 18 | 18 |
| Rui Pereira | 16 | 17 |
| Rita Ferreira | 16 | 16 |

30. A nationwide supermarket retail chain launched an advertising campaign to promote its own brand of mineral water. During two weeks, several advertisements appeared on national TV. Before the campaign started, the company randomly selected 36 of its stores to be part of a study to measure the impact of the campaign. On the Saturday just before the start of the campaign, all these stores reported the number of water bottles of their own brand that were sold during that day. For the Saturday immediately after the end of the campaign, the same count was done for each of the 36 stores. The results obtained led to the conclusion that there was an increase in sales for most of the stores. The average increase in sales per store was 1500 bottles with a standard deviation of 600. What can you conclude regarding the average increase in sales per store for the whole chain of supermarket stores? Use a 95% confidence interval. Can you conclude that the campaign had an effect? Assume a Normal distribution for the population.

31. A random sample of 10 pairs of identical houses was chosen in a large Midwestern city, and a passive solar heating system was installed in one house from each pair. The total fuel bills (in dollars) for three winter months for these homes were then determined, as shown in the accompanying table. Assuming normal population distributions, find a 90% confidence interval for the difference between the two population means.

| Pair | Without passive solar | With passive solar |
|---|---|---|
| 1 | 485 | 452 |
| 2 | 423 | 386 |
| 3 | 515 | 502 |
| 4 | 425 | 376 |
| 5 | 653 | 605 |
| 6 | 386 | 380 |
| 7 | 426 | 395 |
| 8 | 473 | 411 |
| 9 | 454 | 415 |
| 10 | 496 | 441 |

32. We want to know how much do undergraduate students spend on studying on average. Towards this end, we conducted a survey for two independent random samples of students in the UKL and UWL universities. One of the questions that students had to respond was the following: "How many hours do you study per week during the exam period?". The following table presents a summary of the results obtained.

| | UKL students | UWL students |
|---|---|---|
| $n$ | 30 | 40 |
| $\bar{x}$ | 43 | 37 |
| $s^2$ | 90 | 40 |

What is the 95% confidence interval for the difference in average study time of the students in the two universities?

33. According to a psychologist at the Medical College of Virginia, football (soccer) players who "head" the ball 10 or more times a game are at risk for brain damage that could lower their intellectual abilities. In tests involving young male soccer players, those who typically headed the ball 10 or more times a game recorded a mean IQ score of 103, compared to a mean of 112 for those who usually headed the ball once or less a game. Assuming 30 players in each sample, with sample standard deviations of 10 and 8 IQ points for the "headers" and "nonheaders," respectively, construct a confidence interval for the difference in average IQs. Can you conclude that frequent "heading" of the ball by soccer players lowers intellectual performance? Source: Marilyn Elias, "Heading Soccer Ball Can Lower IQ," USA Today, August 14, 1995, p. 1D.

34. In an European country, since 2010, all electricity costumers faced a fixed price tariff of 12 cents/kWh at any period of the day. In 2012, the electricity distribution company in that country conducted an experiment in order to study the impact in the consumption of electricity of introducing a dual tariff. In order to perform this experiment, the company created a group of residential customers that were selected randomly from the population. In this group, called the test group, all customers received information regarding their new dual prices for 2012. None of these costumers declined to participate in the experiment. During the year of 2012, all customers in the test group changed their tariff to the new dual tariff: in peak periods (weekly days between 7pm to 8pm) the price was 15 cents/kWh, while in the other periods of the day the price was 10 cents/kWh. For each costumer in this test group, the company registered their consumption in 2012, as well as in 2011 when all customers faced a single price at any period of the day.

    The company also collected the consumption in peak hours for 2011 and for 2012 for another random sample of 89 costumers, denoted as the control group, that paid the fixed price of 12 cents/kWh at any period of the day in both years.

    In the table below we present some descriptive statistics regarding the change in electricity consumption during peak hours (unit: kWh/day) from 2011 to 2012 for the two groups (note: a negative change corresponds to a reduction in consumption).

| | Control group | Test group |
|---|---|---|
| $\bar{x}$ | -0.20 | -0.27 |
| $\tilde{x}$ | -0.23 | -0.29 |
| $s$ | 0.79 | 0.71 |
| Max | 1.51 | 1.27 |
| Min | -2.00 | -2.03 |
| $n$ | 89 | 100 |

    Calculate a confidence interval, with a confidence level of 95%, for the difference in means between the two groups (test and control groups). Comment the results. What is the conclusion?

35. Opinion polls are widely used by politicians all over the world as a way to monitor voters' support. Six months ago, a poll was conducted to determine the support for the leader of a political party. From a random sample of size 400, 54% of the respondents said they supported that politician. During this month, another study with 900 respondents showed that 45% supports that same leader. Present a 95% confidence interval for the change in the percentage of voters supporting the politician. From that result, can we infer that voter's support went down?

36. The marketing director of a bank wants to study the impact of a new brochure in terms of attracting the attention of its clients to the services provided by personal account managers. The new brochures were mailed to a random sample of 200 clients. The old brochure was sent to another random sample of 100 clients. Of the 200 clients that received the new brochure, 90 contacted their personal account managers during following month. Of the other 100 clients, only 40 made a contact. What would have been the additional impact of the new brochure, relative to the old one, if it was sent to all bank clients? Present a point estimator for the difference in proportions and the corresponding 95% confidence interval. What can you conclude?

37. A recent study reveals that 80% of women and 70% of men are regular users of social networks. Knowing that this study was based on 32 women and 42 men, construct a 95% confidence interval for the difference between the population proportions for men and women. Interpret.

38. (Newbold, p. 325) The President's policy on domestic affairs received a 45% approval rating in a recent poll. The margin of error was given as 0.035. What sample size was used for this poll if we assume a 95% confidence level?

39. (Newbold, p. 340) Suppose that for a random sample of 200 firms that revalued their fixed assets, the mean ratio of debt to tangible assets was 0.517 and the sample standard deviation was 0.148. For an independent random sample of 400 firms that did not revalue their fixed assets, the mean ratio of debt to tangible assets was 0.489 and the sample standard deviation was 0.158. Find a 99% confidence interval for the difference between the two population means.

40. (Newbold, p. 342) Supermarket shoppers were observed and questioned immediately after putting an item in their cart. Of a random sample of 510 choosing a product at the regular price, 320 claimed to check the price before putting the item in their cart. Of an independent random sample of 332 choosing a product at a special price, 200 made this claim. Find a 90% confidence interval for the difference between the two population proportions.

41. Solve the "Time-of-use tariffs - Part 1" case study.

# 5 Hypothesis testing

1. An hypothesis test with a significance level of 5% means that:

   (a) The null hypothesis will be rejected with a 5% probability.

   (b) If the null hypothesis is correct, the probability that the test leads to its rejection is equal to 5%.

   (c) If the null hypothesis is correct, the probability that the test leads to its rejection is equal to 95%.

   (d) If this test is applied many times, then in 5% of the cases the null hypothesis will not be correct.

2. One statistical study reported that "the p-value of our test statistic equals 0.03". What can we conclude?

   (a) Their null hypothesis is not rejected for $\alpha = 10\%$

   (b) Their null hypothesis is not rejected for $\alpha = 5\%$

   (c) Their null hypothesis is not rejected for $\alpha = 1\%$

   (d) Their null hypothesis is rejected for $\alpha = 1\%$ provided they have a two-tail test.

3. The quality department in a lamp manufacturing plant wants to know if the mean duration of a lot of 10 thousand lamps produced with a new technology exceeds the 2000 hours benchmark guaranteed by a standard technology. The trials conducted with a random sample of 100 lamps from that lot resulted in an average duration of 2100 hours. Suppose that the standard deviation for the whole lot of 10 000 lamps was equal to 500 hours.

   (a) Define the null and alternative hypothesis.

   (b) For a significance level of 5%, is the statistical evidence enough to conclude that the mean duration of the lamps in that lot is above 2000 hours? Be clear about the test statistic being used, its distribution if the null hypothesis is true, the rejection region, and the interpretation of the test results.

   (c) If the standard deviation of the lamp durations in that lot was 1000, instead of 500, how would the answers to the previous questions change? What can you conclude?

   (d) If the significance level was 1%, instead of 5%, how would the answers to the previous questions change? Interpret the result?

4. The company that provides piped water for consumption in the city Lusa is required by law to meet certain quality standards. In particular, the average level of nitrates should not exceed the value of 50 (mg/L NO3). Suppose that the distribution of nitrate levels of the water reaching the consumer homes follows a normal distribution with a certain mean and variance. In the face of some doubts about whether the company is fulfilling the criterion legally defined for nitrates, a consumer association decided to conduct a study on this issue. They collected a sample of water from the taps of 8 randomly chosen consumers and measured the respective levels of nitrates. When the results arrive, they will conduct a statistical hypothesis test in order to conclude something about this problem. A significance level of 2.5% must be used. Suppose that the variance of the levels of nitrates in the population is known and equal to 36.

   (a) What is the null hypothesis?

   (b) What is the alternative hypothesis?

   (c) What test statistic should be used to conduct the hypothesis test? Indicate the formula, defining all symbols that you use. Also state its distribution if $H_0$ is true.

   (d) What decision rule should be used?

(e) The results just arrived. The 8 observations of the nitrate level have an average equal to 56 and a variance of 32.

   i. What is the value of the test statistic?
   ii. What is the conclusion of the hypothesis test?
   iii. Obtain the p-value of this test?
   iv. For what significance levels would the conclusion of the test be different from that obtained in ii.?

(f) Finally, suppose the variance of the nitrate levels in the population was not known. Answer all the previous questions.

5. EDI is the electrical utility company in Highland that is responsible for all the distribution of electrical energy to residential consumers in the country. During 2010, the company launched a major energy saving program directed at all consumers in Highland. A sample of 100 customers was selected at random from the population. For each customer, the company calculated the variation in electricity consumption in 2010 relative to the previous year (in kWh/day). Next, we present some descriptive statistics for this sample:

   - Min. = -2.5
   - Max. = 1.5
   - Average = -0.5
   - Median = -0.5
   - Standard Deviation = 2

   We want to check if on average, the customers in Highland reduced their electricity consumption in 2010, relative to the previous year, by more than 0.3 kWh/day. Test this hypothesis using a statistical approach. Be clear about: (i) null and alternative hypothesis, (ii) test statistic and its distribution, (iii) significance level to use, (iv) critical value, (v) decision rule, (vi) p-value, and (vii) final conclusion. What can you conclude about the impact of the energy saving program?

6. To study the number of hours that college students sleep on average, we selected a random sample of 25 students. Below, are presented some descriptive statistics regarding the number of hours each student slept the night before:

   - Min. = 5.8
   - Max. = 9.5
   - Average = 7.0
   - Median = 6.5
   - Standard Deviation = 1.0

   (a) Assuming that the standard deviation for the population equals 0.5, test the hypothesis that during the night before, students slept on average 7.5 hours, using a significance level of 5%. Be clear about all steps and assumptions that are required to conduct the test.
   (b) Solve the previous questions if it is not possible to assume a specific value for the population standard deviation.
   (c) For each of the previous questions, calculate the p-value and indicate its interpretation.
   (d) What is the relationship between the test you conducted in (a) and (b), and a confidence interval for the mean?

7. A company is planning on introducing in all its plants a 20 minute break through the morning shift so that employees can perform relaxation exercises. Before deciding that, the company did an experiment. For a random sample of 10 employees they measured productivity in the week before the break was implemented and in the week after the break was implemented. The results obtained are presented in the following table.

| Pair | Before | After |
|---|---|---|
| 1 | 7 | 9 |
| 2 | 6 | 7 |
| 3 | 8 | 9 |
| 4 | 5 | 7 |
| 5 | 4 | 6 |
| 6 | 7 | 8 |
| 7 | 8 | 7 |
| 8 | 5 | 6 |
| 9 | 9 | 9 |
| 10 | 6 | 7 |

What statistical hypothesis test can you use to check if the break leads to an increase in productivity and what is the conclusion? What assumptions are necessary for this test to be valid? Based on the results obtained, would you recommend implementing that 20 minute break for all the employees?

8. We want to study a random variable $X$ with a normal distribution in the population. For this purpose, a sample with four observations will be used. The observations in this sample are independent and all have the same expected value. Suppose the variance of each observation in the sample is increasing. Specifically, the variance of the $1^{st}$ observation equals a certain value $\sigma^2$, the variance of the $2^{nd}$ observation is equal to twice the variance of the $1^{st}$ observation, the variance of the $3^{rd}$ observation is three times the variance of the $1^{st}$ observation, and the variance of the $4^{th}$ observation is four times the variance of the 1st observation. Answer the following questions by presenting and justifying all the necessary steps. When necessary, use a significance level of 5%. Assume that the expected value is unknown but the variance of the first observation in the sample is known and equal to 1. We want to test if the expected value is equal to 50 or not. Propose a methodology to perform a test of statistical hypotheses. Explain. Apply your proposed test procedure to the case of a specific sample for which the sample mean was equal to 53. Indicate whether the null hypothesis is rejected or not. Present and justify all steps.

9. A random sample is going to be obtained from a certain population with a Normal distribution with variance equal to 2.5. Using that sample, one wants to test whether the population mean is or is not equal to zero. Although the observations in the sample have the same distribution, they are not independent. The correlation between them is equal to 0.5. For a particular sample of four observations, the resulting sample mean was 1. Propose a procedure to implement a statistical hypothesis test. Is the null hypothesis rejected? Assume the observations have a joint Normal distribution.

10. (Newbold, p. 362) A pharmaceutical manufacturer is concerned that the impurity concentration in pills should not exceed 3%. It is known that from a particular production run impurity concentrations follow a normal distribution with a standard deviation of 0.4%. A random sample of 64 pills from a production run was checked, and the sample mean impurity concentration was found to be 3.07%.

    (a) Test at the 5% level the null hypothesis that the population mean impurity concentration is 3% against the alternative that it is more than 3%.

    (b) Find the p-value for this test.

    (c) Suppose that the alternative hypothesis had been two-sided, rather than one-sided, with the null hypothesis $H_0 : \mu = 3$. State, without doing the calculations, whether the p-value of the test would be higher than, lower than, or the same as that found in part (b). Sketch a graph to illustrate your reasoning.

    (d) In the context of this problem, explain why a one-sided alternative hypothesis is more appropriate than a two-sided alternative.

11. (Newbold, p. 364) Test the hypotheses

$$H_0 : \mu \leq 100 \ \ vs \ \ H_1 : \mu > 100$$

using a random sample of size $n = 25$, a probability of type I error equal to 0.05, and the following sample statistics. Assume a normal population.

(a) $\bar{x} = 106; s = 15$

(b) $\bar{x} = 104; s = 10$

(c) $\bar{x} = 95; s = 10$

(d) $\bar{x} = 92; s = 18$

12. (Newbold, p. 365) You have been asked to evaluate single-employer plans after the establishment of the Health Benefit Guarantee Corporation. A random sample of 76 percentage changes in promised health benefits was observed. The sample mean percentage change was 0.078, and the sample standard deviation was 0.201. Find and interpret the p-value of a test of the null hypothesis that the population mean percentage change is 0 against a two-sided alternative.

13. (Newbold, p. 365) In contract negotiations a company claims that a new incentive scheme has resulted in average weekly earnings of at least \$400 for all customer service workers. A union representative takes a random sample of 15 workers and finds that their weekly earnings have an average of \$381.35 and a standard deviation of \$48.60. Assume a normal distribution.

(a) Test the company's claim.

(b) If the same sample results had been obtained from a random sample of 50 employees, could the company's claim be rejected at a lower significance level than that used in part (a)?

14. (Newbold, p 379) An insurance company employs agents on a commission basis. It claims that in their first-year agents will earn a mean commission of at least \$40,000 and that the population standard deviation is no more than \$6,000. A random sample of nine agents found for commission in the first year,

$$\sum_{i=1}^{9} x_i = 333 \ \text{ and } \ \sum_{i=1}^{9} (x_i - \bar{x})^2 = 312$$

where $x_i$ is measured in thousands of dollars and the population distribution can be assumed to be normal. Test, at the 5% level, the null hypothesis that the population mean is at least \$40,000.

15. Last week, a market research company conducted an opinion survey about the attitudes of Portuguese drivers. They used a random sample of 100 drivers. One of the questions included in the questionnaire was: "Are you a better driver than a median Portuguese driver?". Each respondent answered "Yes" or "No". From the random sample of 100 drivers, 70 answered "Yes." Mr. Ferreira believes that 50% of the Portuguese drivers "think they are better drivers that a median Portuguese driver". What can you conclude about that belief? Use a statistical hypothesis test that has a 95% probability of not leading to a wrong conclusion if Mr. Ferreira is indeed correct. Be clear as to: (i) null and alternative hypotheses, (ii) test statistic and its distribution, (iii) level of significance to be used, (iv) critical value, (v) decision rule and (vi) final conclusion. What is the p-value of the test? How is the result of the hypothesis test related to a confidence interval for the proportion?

16. A credit card company plans to offer a special incentive program to customers who charge at least \$500 next month in their credit cards. The marketing department has pulled a sample of 500 customers from the same month last year and noted that the mean amount charged was \$478.19 and the median amount was \$216.48. The finance department says that the only relevant

quantity is the proportion of customers who spend at least \$500. If that proportion is not more than 25%, the program will lose money. Among the 500 customers, 148 or 29.6% of them charged \$500 or more. Use an hypothesis test approach to solve this problem. Can we use a confidence interval to test whether the goal of 25% for all customers was met? Note: One-sided hypothesis tests may not lead to the same conclusions as confidence intervals, which are inherently two-sided. In addition, the standard error differs across the two procedures: hypothesis tests use $p = 0.25$, whereas confidence intervals use $\hat{p}$.

17. "Browser Wars" is a metaphorical term used to describe periods of fierce competition for dominance in market share in the use of web browsers. The term was first used in the second half of the 90s when Microsoft's Internet Explorer dethroned the then leader Netscape Navigator. There are several ways to measure market shares. One is to conduct surveys among random samples of the population. Based on a survey conducted in April 2012 for a random sample of 100 individuals of the Portuguese population with internet access it was found that 45 of them used Internet Explorer as the main browser. Based on the results of this survey answer the following questions

    (a) Google states that in Portugal, Internet Explorer lost market leadership in April with a market share below 50%. Test this hypothesis based on the results of the survey conducted in Portugal. Specify the null and alternative hypotheses, the formula for the test statistic and its distribution under the null hypothesis, the level of significance and the respective critical value, and the decision rule. What is the final conclusion of the test?

    (b) In France, it is suspected that in June the market share of Internet Explorer will be less than 50%. What should be the size of the sample to be collected from the French population in order for the statistical test of this hypothesis, with a significance level of 5%, to be able to confirm this suspicion with a probability of 97.5% if in fact the share of Internet Explorer in France was equal to 20% in the month of June?

18. (Newbod, p. 368) A random sample of women is obtained, and each person in the sample is asked if she would purchase a new shoe model. The new shoe model would be successful in meeting corporate profit objective if more than 25% of the women in the population would purchase this shoe model. The following hypothesis test can be performed at a level of $\alpha = 0.03$ using $\hat{p}$ as the sample proportion of women who said yes.

    $$H_0 : p \leq 0.25 \;\; vs \;\; H_1 : p > 0.25$$

    What value of the sample proportion, $\hat{p}$, is required to reject the null hypothesis, given the following sample sizes?

    (a) $n = 400$
    (b) $n = 225$
    (c) $n = 625$
    (d) $n = 900$

19. (Newbold, p. 368) In a random sample of 361 owners of small businesses that had gone into bankruptcy, 105 reported conducting no marketing studies prior to opening the business. Test the hypothesis that at most 25% of all members of this population conducted no marketing studies before opening their businesses. Use $\alpha = 0.05$.

20. (Newbold, p. 368) Of a random sample of 199 auditors, 104 indicated some measure of agreement with this statement: Cash flow is an important indication of profitability. Test at the 10% significance level against a two-sided alternative the null hypothesis that one-half of the members of this population would agree with this statement. Also find and interpret the p-value of this test.

21. (Newbold, p. 379) A random sample of 104 marketing vice presidents from large Fortune 500 corporations was questioned on future developments in the business environment. Of those sample members, 50 indicated some measurement of agreement with this statement: Firms will concentrate their efforts more on cash flow than on profits. What is the lowest level of significance at which the null hypothesis, which states that the true proportion of all such executives who would agree with this statement is one-half, can be rejected against a two-sided alternative?

22. An automobile insurance company is worried about the high repair costs charged by a reputed auto repair shop, "CarShock". To assess whether this shop is charging too much for their repairs, they selected a random sample of 16 vehicles needing repairs due to car accidents. Each vehicle was taken both to "CarShock" and to another trusted repair shop (TRS). Each shop provided an estimate of the repair cost for each vehicle. The following estimates (in euros) were obtained:

| Vehicle | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| CarShock | 1000 | 2000 | 1500 | 2500 | 3000 | 2500 | 2200 | 1500 |
| TRS | 900 | 2000 | 1400 | 2600 | 2000 | 2500 | 2200 | 1600 |
| Vehicle | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| CarShock | 2500 | 1900 | 2000 | 2500 | 3000 | 1600 | 2200 | 2900 |
| TRS | 2700 | 1400 | 1800 | 3000 | 2900 | 1300 | 2000 | 2900 |

(a) Using a statistical hypothesis test, what can you conclude? Be clear about all steps and necessary assumptions.

(b) What is the p-value and its interpretation?

23. Consider the following article published in The New York Times (July 20, 2009).

Drilling Down

**Snack Ads Spur Children to Eat More**

By ALEX MINDLIN

Psychologists recognize that certain behaviors can be automatic. For example, unrecognized external stimuli can unconsciously stir us to anger, spur us to loyalty or incite us to rudeness without our knowing it. A new study finds that seeing food ads on television can induce people to eat more snacks while watching.

In one experiment, conducted by researchers from the Rudd Center for Food Policy and Obesity at Yale, 118 children, ages 7 to 11, were each given bowls of Goldfish crackers and then left to watch a 14-minute cartoon. During the commercial breaks, some of the children saw ads for games and entertainment; others watched four spots for unhealthy snacks like waffle sticks with syrup, fruit roll-ups and potato chips. The children who saw the food spots ate 45 percent more Goldfish (mean=28.5 grams) than those who watched the game commercials (mean=19.7 grams).

According to the authors, the experiment suggests "a direct causal link between food advertising and greater snack consumption."

That runs counter to the industry line, which has long held that food ads merely promote competition among similar brands, rather than inspiring extra food consumption or inducing unhealthier food preferences.

Although not reported, suppose the sample size in each group was the same, and the sample standard deviation for those children that watched the food ad was 9 grams and for those that did not watch the food ad was 8 grams. What can you conclude? Perform a hypothesis test using the confidence interval approach.

24. We want to know how much do undergraduate students spend on studying on average. Towards this end, we conducted a survey for two independent random samples of students in the UKL and UWL universities. One of the questions that students had to answer was the following: "How many hours do you study per week during the exam period?". The following table presents a summary of the results obtained.

| | UKL students | UWL students |
|---|---|---|
| $n$ | 30 | 40 |
| $\bar{x}$ | 43 | 37 |
| $s^2$ | 90 | 40 |

On average, do students in these two universities spend the same time studying? Perform a statistical hypothesis test by analyzing the results of a confidence interval for the difference in means.

25. A group of marketing researchers conducted an experiment to analyze the memory of children when exposed to television ads with or without sound. The target population consists of children aged 6 years. Two random samples of this population were selected independently of each other. Every child in the first sample was shown a 1 minute ad for a brand of Honey Cereals with image and sound. Children in the second sample were shown the same ad but without sound. In the end, each child was asked about 50 items that appeared in the ad. The number of items that each child remembered correctly was recorded. A summary of results is presented in the following table.

| | With sound | Without sound |
|---|---|---|
| $n$ | 30 | 40 |
| $\bar{x}$ | 33 | 36 |
| $s^2$ | 15 | 20 |

(a) Is a child's memory affected by sound? Present an accurate and complete response using a statistical approach, not forgetting to mention the hypotheses to be tested. Comment the results.

(b) In another study, using the same format as above, but with 10 year olds, the only information available about the results was that the p-value of the statistical test was 0.004. Using only this information, is it possible to conclude anything about the memory of 10 year olds? If yes, what can you conclude and why? If not, please explain why and what information is missing.

26. Opinion polls are widely used by politicians all over the world as a way to monitor voters support. Six months ago, a poll was conducted to determine the support for the leader of a political party. From a random sample of size 400, 54% of the respondents said they supported that politician. During this month, another study with 900 respondents showed that 45% supports that same leader.

(a) Present a 95% confidence interval for the change in the percentage of voters supporting the politician.

(b) Use the answer to the previous question to perform a statistical hypothesis test to check if the popularity of that politician remained constant or not.

27. A market research company conducts every month an opinion survey directed at entrepreneurs regarding their future economic expectations. One of the questions is the following: "Do you expect to have a reduction in your company's economic activity during the next month?". The December survey was based on a random sample of 40 entrepreneurs, of which 30 stated that they expected a reduction in their economic activity. In the November survey, the sample included 50

entrepreneurs and the result was that 25 of them expected to have a reduction in their economic activity. Suppose this sample was obtained independently from the December sample. Check if the proportion of Portuguese entrepreneurs expecting a reduction in their economic activity remained stable from November to December. Perform the test using a confidence level of 95% for the difference in proportions.

28. Does music training improve school performance? To answer this question, a recent study carried out in Ireland was based on a sample of 100 randomly selected individuals from the population of 18 years old residents that during their school career had some sort of music training. The results indicate that 30% of these 100 students had already entered the university.

    (a) The benchmark percentage for OECD countries is 20%. Test the hypothesis that the percentage in Ireland is higher than the benchmark value for OECD. Be clear about all the details necessary to conduct such statistical test. What is the conclusion?

    (b) You need to conduct a similar study in Portugal in order to test the hypothesis that the percentage is higher than the benchmark value of 20%. What should be the sample size so that you can reject the null hypothesis, at a significance level of 5%, with a 99% probability in case the percentage is actually higher than the OECD benchmark by 10 percentage points? Provide all the necessary calculations.

    (c) In the case of Ireland, it was also analyzed a sample of 200 individuals randomly selected from all 18 years old residents that during their school career never had any music training. Only 10% of those 200 students had already entered the university. Can one conclude that these percentages differ in the two populations (18-year old that had music training vs. no music training)? Be clear about all the details required to perform such a test.

    (d) Does the conclusion from the statistical test in the previous question allow one to answer the question: Does music education improve academic performance in Ireland? How and under what conditions would it be possible to conclude anything about a causal relationship in this particular problem?

29. Solve the "Time-of-use tariffs - Part 2" case study.

# 6 Linear regression

1. (Newbold, p. 420) Complete the following for the $(x, y)$ pairs of data points: (1, 5), (3,7), (4, 6), (5,8) and (7,9).

   (a) Prepare a scatter plot of these data points.

   (b) Compute $b_1$.

   (c) Compute $b_0$.

   (d) What is the equation of the regression line?

2. (Newbold, p. 420) The following data give $X$, the price charged per piece of plywood, in dollars, and $Y$, the quantity sold (in thousands). Data available in the *Excel* file *Plywood.xlsx*.

| Price per piece ($X$) | Thousands of pieces sold ($Y$) |
|---|---|
| 6 | 80 |
| 7 | 60 |
| 8 | 70 |
| 9 | 40 |
| 10 | 0 |

   (a) Prepare a scatter plot of these data points.

   (b) Compute the covariance.

   (c) Compute and interpret $b_1$.

   (d) Compute $b_0$.

   (e) What quantity of plywood would you expect to sell if the price were \$7 per piece?

3. (Newbold, p. 421) A random sample of data for 7 days of operation produced the following (price, quantity) data values(data available in the *Excel* file *Paint.xlsx*):

| Price per gallon of paint ($X$) | Quantity sold ($Y$) |
|---|---|
| 10 | 100 |
| 8 | 120 |
| 5 | 200 |
| 4 | 200 |
| 10 | 90 |
| 7 | 110 |
| 6 | 150 |

   (a) Prepare a scatter plot of the data.

   (b) Compute and interpret $b_1$.

   (c) Compute and interpret $b_0$.

   (d) How many gallons of paint would you expect to sell if the price is \$7 per gallon?

4. (Newbold, p. 421) A large consumer goods company has been studying the effect of advertising on total profits. As part of this study, data on advertising expenditures and total sales were collected for a five-month period and are as follows: (10,100), (15,200), (7,80), (12, 120) and (14, 150). The first number is advertising expenditures and the second is total sales. Data available in the *Excel* file *Advertising.xlsx*.

   (a) Plot the data.

   (b) Does the plot provide evidence that advertising has a positive effect on sales?

(c) Compute the regression coefficients, $b_1$ and $b_0$.

5. (Newbold, p. 421) Abdul Hassan, president of Floor Coverings Unlimited, has asked you to study the relationship between market price and the tons of rugs supplied by his competitor, Best Floor, Inc. He supplies you with the following observations of price per ton and number of tons, obtained from his secret files: (2,5), (4,10), (3,8), (6,18), (3,6), (5,15), (6,20) and (2,4). The first number for each observation is price and the second is quantity. Data available in the *Excel* file *Rugs.xlsx*.

   (a) Prepare a scatter plot.

   (b) Determine the regression coefficients, $b_0$ and $b_1$.

   (c) Write a short explanation of the regression equation that tells Abdul how the equation can be used to describe his competition. Include an indication of the range over which the equation can be applied.

6. (Newbold, p. 421) The following ordered pairs provide data about some Nestlé snacks, where the first number is grams of sugar and the second is the number of calories for each snack: (3,110), (14, 180), (13, 150), (11,120), (8, 100), (5,70), (7,140), (15, 200) and (12, 130). Data available in the *Excel* file *Nestlé.xlsx*.

   (a) Construct a scatter plot of the data. Does a clear linear relationship exist between the two variables?

   (b) Estimate the regression equation and identify the value of the slope.

   (c) Which conclusion can you draw from your results?

7. Consider Example 11.1 in page 420 in Newbold et al. and answer the following questions.

**Example 11.1 Manufacturing Plant (Regression Line)**

The Rising Hills Manufacturing Company in Redwood Falls regularly collects data to monitor its operations. These data are stored in the data file **Rising Hills**. The number of workers, $X$, and the number of tables, $Y$, produced per hour for a sample of 10 workers is shown in Figure 11.1. If management decides to employ 25 workers, estimate the expected number of tables that are likely to be produced.

**Solution** Using the data, we computed the descriptive statistics:

$$Cov(x, y) = 106.93,\ s_x^2 = 42.01,\ \bar{y} = 41.2,\ \bar{x} = 21.3$$

From the covariance we see that the direction of the relationship is *positive*.

Using the descriptive statistics, we compute the sample regression coefficients:

$$b_1 = \frac{Cov(x, y)}{s_x^2} = \frac{106.93}{42.01} = 2.545$$

$$b_0 = \bar{y} - b_1\bar{x} = 41.2 - 2.545(21.3) = -13.02$$

From this, the sample regression line is as follows:

$$\hat{y} = b_0 + b_1 x = -13.02 + 2.545x$$

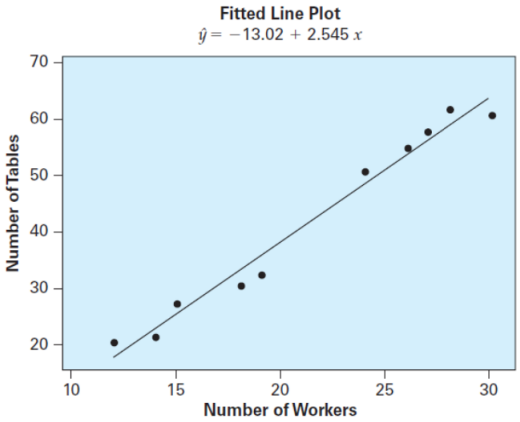
For 25 employees we would expect to produce

$$\hat{y} = -13.02 + 2.545(25) = 50.605$$

or approximately 51 tables. In most situations we use a statistical software package such as Minitab or a spreadsheet such as Excel to obtain the regression coefficients to reduce the work load and improve computational accuracy.

Because the number of workers in the Rising Hill Manufacturing Plant ranged from 11 to 30, we cannot predict the number of tables produced per hour if 100 workers were employed.

**Figure 11.1**




(a) Interpret Figure 11.1 above.

(b) Solve Example 11.1 above using Excel: you should confirm the solution that is presented in the book. The sample data for this example is available in the *Excel* file *Rising Hills.xlsx*.

(c) In Example 11.1 a linear regression model was developed. Use that model to answer the following:

   i. Interpret the coefficient $b_1 = 2.545$ for the plant manager.

   ii. How many tables would be produced on average with 19 workers?

   iii. Suppose you were asked to estimate the number of tables produced if only five workers were available. Discuss your response to this request.

(d) What is the difference between a population linear model and an estimated linear regression model? In addition, explain the difference between the residual $e_i$ and the model error $\varepsilon_i$.

(e) Calculate with *Excel:* i) the sample variance of Y, and ii) the coefficient of determination $R^2$. Interpret these values.

8. (Adapted from Basic Business Statistics - Concepts and Apps 12e - Berenson et. al. 2012) A company that holds the DVD distribution rights to movies previously released only in theatres wants to estimate sales revenue of DVDs based on box office success. The box office gross revenues (in \$millions) for each of 22 movies in the year that they were released and the DVD revenue (in \$millions) can be found in the *Excel* file *Movie.xls*).

   (a) Assuming a linear relationship between these variables, what should be considered as the dependent and the independent variable?

   (b) For these data use Excel to:

      i. construct a scatter plot;

      ii. use the least-squares method to estimate the regression coefficients;

      iii. interpret the intercept and slope, in this problem, and explain the difference between the least-squares estimates and the corresponding population values;

      iv. obtain the residuals, plot them, and identify any particular large ones;

      v. interpret the residuals and explain the difference between these residuals and the population random error terms;

      vi. predict the DVD sales revenue for a movie that had a box office gross revenue of \$75 million.

9. (Adapted from The Practice of Statistics for Business and Economics by Moore et al. 2011) Nash Information Services provides information and analytical services to the movie industry, including statistical models for predicting movie revenue. Consider a simple random sample of 40 movies released over a five-year period. This sample was collected to see if information available soon after the theatrical release can successfully predict total box office revenue. The larger population of interest consists of all movies released over this five-year period. The response variable is a movie's total U.S. box office revenue (USRevenue). Among the explanatory variables are the movie's budget (Budget), opening week revenue (Opening), and how many theaters the movie was in for the opening week (Theaters). All dollar amounts are measured in millions of U.S. dollars. The data are in the *Excel* file *Box office.xls*.

   (a) Estimate a linear regression model in Excel that can be used to predict total U.S. box office revenue.

   (b) Interpret the estimated coefficients. Explain the difference between the least-squares estimates and the corresponding population values.

   (c) Compute the residuals and plot them. Interpret the residuals and explain the difference between these residuals and the population random error terms.

   (d) Are there any particularly large residuals? Does the sign of these large residuals, either positive or negative, convey any interesting information? How do you interpret them?

   (e) In the first panel of the Excel output appears a Standard Error of 41.8. What is that?

   (f) What is the meaning of the R Square?

   (g) What conclusion can you take by analyzing the estimated t-statistics? And the p-values?

   (h) Do the confidence intervals presented confirm the t-statistics results?

   (i) Using the reported coefficient estimates and standard errors, compute 90% confidence intervals.

   (j) Re-estimate the model dropping the least significant explanatory variable. Are all the variables in this new model significant now?

   (k) For the best model, predict total U.S. box office revenues for a movie with a budget of \$100 million that earns \$10 million in 3000 theaters in its first week.

10. (Newbold, p. 481) An aircraft company wanted to predict the number of worker-hours necessary to finish the design of a new plane. Relevant explanatory variables were thought to be the plane's top speed, its weight, and the number of parts it had in common with other models built by the company. A sample of 27 of the company's planes was taken, and the following model was estimated:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i$$

where,

- $y_i$ - design effort, in millions of worker-hours
- $x_{1i}$ - plane's top speed, in miles per hour
- $x_{2i}$ - plane's weight, in tons
- $x_{3i}$ - percentage number of parts in common with other models

The estimated regression coefficients were as follows: $b_0 = 2, b_1 = 0.661, b_2 = 0.065$ and $b_3 = -0.018$. Interpret these estimates.

11. (Newbold, p. 481) In a study of the influence of financial institutions on bond interest rates in Germany, quarterly data over a period of 12 years were analyzed. The postulated model was

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$$

where,

- $y_i$ - change over the quarter in the bond interest rates
- $x_{1i}$ - change over the quarter in bond purchases by financial institutions
- $x_{2i}$ - change over the quarter in bond sales by financial institutions

The estimated regression coefficients were as follows: $b_1 = 0.057$ and $b_2 = -0.065$. Interpret these estimates.

12. (Newbold, p. 481) The following model was fitted to a sample of 30 families in order to explain household milk consumption:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$$

where,

- $y_i$ - milk consumption, in quarts per week
- $x_{1i}$ - weekly income, in hundreds of dollars
- $x_{2i}$ - family size

The least squares estimates of the regression parameters were as follows: $b_0 = -0.025, b_1 = 0.052$ and $b_2 = 1.14$.

(a) Interpret the estimates $b_1$ and $b_2$.

(b) Is it possible to provide a meaningful interpretation of the estimate $b_0$?

13. Consider the "Savings and Loan Associations" example in page 478 in Newbold et. al. The dataset used for this example is also available in the *Excel* file *Savings and loan data.xls*. Answer the following questions.

(a) Use Excel to estimate the multiple regression model of the profits as a function of the revenue per dollar and of the number of offices.

i. Interpret $b_0$, $b_1$, and $b_2$.

ii. Interpret the $R^2$.

iii. What is the difference between the Standard Error of the regression and the Standard Errors for each regression coefficient?

iv. Find the 99% confidence intervals for the regression parameters.

v. Test the null hypothesis that the slope of $X_1$ equals 0.5 against the alternative that it is different from 0.5 using a t-statistic. Why is that in your Excel output the t-statistic for $X_1$ gives a different value?

(b) Use Excel to:

i. Regress $Y$ on $X_1$.

ii. Regress $Y$ on $X_2$. Explain why the estimated slopes obtained in (i) and (ii) are different from the slopes obtained in the multiple regression. Which ones are preferable?

14. Using a sample of 100 families randomly selected from all over the country, it was collected information to study health expenditures of Portuguese families. For each family, data for the following variables were observed:

- $Y$ - Monthly health expenditures (euros);
- $INC$ - Annual income (hundreds of euros);
- $CHILD$ - Number of children;
- $URB$ - Equal to 1 if family lives in an urban region, equal to 0 otherwise.

Using these data the following regression model was estimated:

$$Y_i = \beta_0 + \beta_1 INC_i + \beta_2 CHILD_i + \beta_3 URB_i + \varepsilon_i$$

The estimation results obtained using Excel were:

SUMMARY OUTPUT

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.41242 |
| R Square | 0.17009 |
| Adjusted R Square | 0.144156 |
| Standard Error | 10.67597 |
| Observations | 100 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *Significance F* |
|---|---|---|---|---|---|
| Regression | 3 | 2242.512 | 747.5039 | 6.558413 | 0.000442 |
| Residual | 96 | 10941.73 | 113.9763 | | |
| Total | 99 | 13184.24 | | | |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* |
|---|---|---|---|---|
| Intercept | 97.39896 | 3.863172 | 25.21217 | 4E-44 |
| INC | 0.048717 | 0.018643 | 2.613147 | 0.010415 |
| CHILD | 39.74978 | 17.96329 | 2.212834 | 0.029279 |
| URB | 4.481669 | 2.337796 | 1.917049 | 0.058205 |

(a) Select the correct statement:

i. The URB variable has a statistically significant impact on average health expenditures for a significance level of $\alpha = 5\%$.

ii. The URB variable has a statistically significant impact on average health expenditures for a significance level of $\alpha = 10\%$.

iii. The URB variable has no statistically significant impact on average health expenditures for a significance level of $\alpha = 5\%$. However if $\alpha = 1\%$ the impact would be statistically significant.

iv. None of the above statements is correct.

(b) For a significance level of 5% which of the following variables are statistically significant?

i. Only INC.
ii. INC and CHILD
iii. Only URB
iv. INC, CHILD and URB.

(c) The 95% confidence interval for the coefficient of the URB variable is given by (numbers are rounded to the nearest integer) is

i. $[0, 9]$
ii. $[1, 8]$
iii. $[2, 7]$
iv. $[3, 6]$

(d) Assuming everything else held constant, by how much is it estimated that health expenditures will vary as a consequence of an increase of 100 euros in annual income?

i. 4.8717 euros per month.
ii. 4.8717%.
iii. 0.048717 euros per month.
iv. 0.048717%.

(e) The above regression is going to be re-estimated excluding the URB variable. Which of the following statements is correct?

i. The new $R^2$ may be less than, equal to, or larger than 0.17009.
ii. The new $R^2$ cannot be above 0.17009.
iii. The new $R^2$ cannot be below 0.17009.
iv. The new $R^2$ will be equal to the adjusted $R^2$.

(f) According to the regression results it is expected that urban families without any children and an annual income of ten thousand euros have monthly health expenditures of:

i. 40 euros.
ii. 107 euros.
iii. 147 euros.
iv. 590 euros.

(g) Test the hypothesis that: "*ceteris paribus,* each additional child on average leads to an additional 50 euros per month in health expenditures". Be clear about null and alternative hypotheses to test, significance level, decision rule, and final decision (reject or not reject the null hypothesis).

(h) Suppose the regression model was re-estimated but excluding the INC variable. Knowing that, in Portugal, urban families have incomes that are on average above those in rural regions, it is reasonable to presume that the new estimated coefficient of the URB variable would become:

i. larger.
ii. closer to zero but still positive.
iii. negative.
iv. unchanged.

15. Several economists have studied the impact of political instability and corruption on the quality of government policies. In particular, Fredriksson and Svensson (Journal of Public Economics, 2003) studied the impact of these two variables in determining the degree of enforcement of environmental laws enacted by governments. To test the conclusions obtained in this study, the following variables were analyzed for a random sample of 30 countries:

- $ENV_LAW$ = national environmental law requirement index on a scale of 0 to 100, where 100 represents the maximum requirements;
- $INC_PC$ = income per capita in thousand euros;
- $OECD$ = 1 if country belongs to the Organisation for Economic Co-operation and Development (OECD), = 0 if not;
- $CORRUP$ = corruption index on a scale of 0 to 6, where 6 represents maximum corruption;
- $INSTAB$ = average number of governmental crises per year;
- $CORRUP * INSTAB$ = mathematical product of the $CORRUP$ and $INSTAB$ variables.

Next, we present the result obtained in Excel of the estimation of a linear regression model in which $ENV_LAW$ is the dependent variable.

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.795144 |
| R Square | 0.632254 |
| Adjusted R Square | 0.55564 |
| Standard Error | 23.1703 |
| Observations | 30 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *Significance F* |
|---|---|---|---|---|---|
| Regression | 5 | 22152.24 | 4430.449 | 8.252482 | 0.000119 |
| Residual | 24 | 12884.7 | 536.8626 | | |
| Total | 29 | 35036.95 | | | |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* |
|---|---|---|---|---|
| Intercept | 285.2705 | 247.8098 | 1.151167 | 0.261004 |
| INC_PC | 11.41992 | 3.0004 | 3.806134 | 0.000859 |
| OECD | 5.74888 | 10.14666 | 0.566579 | 0.576261 |
| CORRUP | -19.7392 | 5.053204 | -3.90627 | 0.000668 |
| INSTAB | -110.85 | 37.18412 | -2.98112 | 0.00649 |
| CORRUP*INSTAB | -29.08025 | 9.877075 | -2.94422 | 0.007082 |

(a) If the regression was re-estimated excluding OECD variable, how would the $R^2$ change?

i. The $R^2$ would not decrease.
ii. The $R^2$ would not increase.
iii. The $R^2$ would not change because the p-value for OECD is above 5%.
iv. The $R^2$ would increase because the t-statistic for OECD is below 1.

(b) Estimate by how much would the national environmental law requirement index change as a result of an increase of income per capita by 1000 euros ceteris paribus?

i. 11.4
ii. 11420
iii. 0.0114
iv. 11.4%

(c) Select the correct statement taking into account the estimation results obtained.

i. An increase in corruption, *ceteris paribus,* leads to less demanding environmental laws, but this effect is less pronounced in countries with greater political instability.

ii. An increase in corruption, *ceteris paribus,* leads to less demanding environmental laws, and this effect is more pronounced in countries with greater political instability.

iii. An increase in corruption, *ceteris paribus,* leads to more demanding environmental laws, but this effect is less pronounced in countries with greater political instability.

iv. An increase in corruption, *ceteris paribus,* leads to more demanding environmental laws, and this effect is more pronounced in countries with greater political instability.

(d) Consider the following statement: "On average, *ceteris paribus,* the level of environmental laws in OECD countries is the same as in other countries in the world." Which of the following comments on this statement is more appropriate in view of the results obtained?

i. I disagree, since the p-value of the OECD variable is less than 5%.

ii. I agree, since the p-value of the variable OECD is higher than 5%.

iii. I disagree, since the p-value of the OECD variable is higher than 5%.

iv. I agree, since the coefficient estimate of the OECD variable is positive.

(e) Which factors seem to explain $ENV_LAW$ considering a significance level of 5%?

i. All factors: $INC_PC$, $OECD$, $CORRUP$, $INSTAB$, and $CORRUP * INSTAB$.

ii. Only $OECD$.

iii. Only $INC_PC$, $CORRUP$, $INSTAB$, and $CORRUP * INSTAB$.

iv. Only $CORRUP$.

(f) In the above results the confidence intervals calculated by Excel were omitted. What is the 95% confidence interval for the coefficient of the OECD variable?

i. $[-15, 27]$

ii. $[-4, 16]$

iii. $[-96, 107]$

iv. $[0, 11.5]$

(g) *Ceteris paribus,* the degree of exigency of environmental laws is certainly lower as the level of natural and environmental resources in each country increases. However, the level of natural and environmental resources in each country was not included as an explanatory variable in the regression estimated above. If the level of these resources is positively correlated with the per capita income of each country then, ceteris paribus, the estimated impact of the per capita income on the $ENV_LAW$ index:

i. is most likely over-estimated.

ii. is most likely correctly estimated.

iii. is most likely under-estimated.

iv. is most likely in the interval $[8.4, 14.4]$ with a confidence level of 95%.

16. A Telemarketing firm operating in Lisbon, Porto and Coimbra, wanted to study what factors influenced the productivity of its employees. In particular, it was especially interested in checking whether sales productivity had increased after some of the employees were exposed to an intensive one-week training programme that took place at the end of the $3^{rd}$ quarter of 2016. Even though $4^{th}$ quarter total sales increased relative to the $3^{rd}$ quarter, there was a considerable dispersion of results when considering the individual employees as well as many other factors that influenced sales. To better understand the variability in productivity across employees and to check if the training programme was successful, a regression analysis was conducted. At the end of the $4^{th}$ quarter of 2016, a random sample of 145 employees was selected and the following information collected for each of them:

- SALES = indicator of an employee's sales productivity (related to the percentage of phone calls that were successful and hours spent on each call),
- EXPERIENCE = employee's previous experience in telemarketing (in years),
- AGE = employee's age (in years),
- PORTO = 1 if the employee operates at Porto's branch, = 0 if not,
- COIMBRA = 1 if the employee operates at Coimbra's branch, = 0 if not,
- TRAINING = 1 if the employee took the intensive training programme, = 0 if not.

The following model was estimated:

SUMMARY OUTPUT

| *Regression Statistics* | |
|---|---|
| R Square | 0.789386 |
| Adjusted R Square | 0.780229 |
| Standard Error | 3.464966 |
| Observations | 145 |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* |
|---|---|---|---|---|
| Intercept | 6.205852 | 4.181162 | 1.484241 | 0.1400 |
| EXPERIENCE | 0.885850 | 0.103290 | 8.576352 | 0.0000 |
| AGE | 0.556730 | 0.226741 | 2.455364 | 0.0153 |
| AGE^2 | -0.009821 | 0.003073 | -3.195392 | 0.0017 |
| PORTO | -1.874340 | 0.704695 | -2.659791 | 0.0087 |
| COIMBRA | -10.36313 | 0.842711 | -12.29736 | 0.0000 |
| TRAINING | 1.695174 | 0.730858 | 2.319429 | 0.0218 |

(a) Compute the 95% confidence interval for the coefficient of EXPERIENCE.

(b) Compute the 90% confidence interval for the coefficient of PORTO.

(c) What is the estimated impact on sales productivity of having one more year of experience, everything else held constant?

  i. 88.585.
  ii. 0.88585.
  iii. 88.585%.
  iv. 8.576352.

(d) Is the TRAINING variable significant?

  i. Not at 5%.
  ii. Yes at 5%.
  iii. Not at 10%.
  iv. Yes at 1%.

(e) What city is estimated to have the highest productivity levels on average, everything else held constant?

  i. Lisbon.
  ii. Porto.
  iii. Coimbra.
  iv. The information available is not enough to provide an answer to this question.

(f) According to the value of $R^2$:

i. The included explanatory variables account for 0.789386% of the variability in sales productivity.
ii. The included explanatory variables account for 78.9386% of the variability in sales productivity.
iii. The variability in sales productivity not accounted for by the explanatory variables included in the model equals 0.789386%.
iv. The variability in sales productivity not accounted for by the explanatory variables included in the model equals 78.9386%.

(g) If all the employees in the sample had taken the training programme then:
i. it would not be possible to obtain an estimate of the impact of the training programme.
ii. the estimated parameter value for the TRAINING variable would be smaller.
iii. the estimated parameter value for the TRAINING variable would be the same.
iv. the estimated parameter value for the TRAINING variable would be higher.

(h) What is the estimated productivity of a 50 years old employee operating in Lisbon without any experience and that did not participate in the training programme?
i. 9.5
ii. 11.2
iii. -2.7
iv. 0.0

(i) According to the estimated model, at what age is productivity maximized, everything else held constant?
i. 44
ii. 39
iii. 28
iv. 57

(j) The effect of the training programme on sales productivity may depend on the employee's previous experience. What alternative regression would be needed to appraise this issue? What values of the parameter estimates would you expect? Why?

(k) From the results of the estimated model, what was the impact of the training programme on productivity? Is it statistically significant? The training programme was supposed to bring an average sales productivity gain of more than 1. What can be said about this claim?

(l) An important factor that contributes to productivity is the remuneration scheme of each employer. For instance, some of the employers have a fixed hourly wage while others earn an extra premium for every successful call they make. In the latter case, employees have an extra incentive that may influence productivity positively. What do you expect to have been the consequence on the estimated impact of training on productivity of having omitted this variable? Is the answer to this question dependent on how employees were selected to participate in the training programme?

17. The success of a retail store depends on many factors. If a company wants to forecast sales of a new store or to decide about the best location for new stores, one approach that could be followed would be to collect data on existing stores and use regression analysis to measure the impact of the important factors that explain sales. Examples of factors that could be used are location variables measuring trade area characteristics (population density, mean household income,...) or the level of competition (such as the number of competing stores in the area) and store characteristics (such as the type of store, floor area, or accessibility). A worldwide retail company decided to study the factors affecting the sales of its stores. A random sample of 145 stores located in many different world cities was selected. Data for the year of 2017 were then collected for each of the following variables for each store:

- SALES = total sales in thousand euros,
- INCOME = urban per capita income in thousand euros,
- DENSITY = urban population density,
- COMPET = number of competing stores in the same urban area,
- AREA = store floor area in square meters,
- CENTER = 1 if the store is located within 1 km from the city center, = 0 if not,
- CLOSE = 1 if the store is between 1 km and 5km of the city center, = 0 if not,
- FAR = 1 if the store is more than 5km away from the city center, = 0 if not,
- SHOPPING = 1 if the store is located inside a shopping mall, = 0 if not.

Using SALES as the dependent variable, a linear regression model was estimated by the least-squares method and the following results were obtained:

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.970 |
| R Square | 0.940 |
| Adjusted R Square | 0.936 |
| Standard Error | 0.359 |
| Observations | 145 |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* | *Lower 95%* | *Upper 95%* |
|---|---|---|---|---|---|---|
| Intercept | -1.96 | 1.66 | -1.18 | 0.24 | -5.24 | 1.32 |
| INCOME | 0.64 | 0.21 | 3.05 | 0.00 | 0.23 | 1.06 |
| DENSITY | 0.30 | 0.15 | 1.94 | 0.05 | -0.01 | 0.60 |
| COMPET | -0.50 | 0.10 | -5.12 | 0.00 | -0.69 | -0.31 |
| AREA | 0.46 | 0.32 | 1.46 | 0.15 | -0.17 | 1.09 |
| CLOSE | -0.23 | 0.03 | -8.87 | 0.00 | -0.28 | -0.18 |
| FAR | -0.59 | 0.23 | -2.63 | 0.01 | -1.04 | -0.15 |
| SHOPPING | 0.49 | 0.19 | 2.61 | 0.01 | 0.12 | 0.86 |
| CLOSE* SHOPPING | 0.04 | 0.02 | 2.21 | 0.03 | 0.00 | 0.08 |
| FAR* SHOPPING | 0.53 | 0.13 | 3.96 | 0.00 | 0.26 | 0.79 |

(a) It is possible to say that the variable AREA has a significant impact on sales if the significance level $\alpha$ is such that:

i. $\alpha > 15\%$.
ii. $\alpha < 15\%$.
iii. $\alpha = 5\%$.
iv. $\alpha < 5\%$.

(b) If the impact of a store being located inside a shopping mall is higher for smaller stores (stores with smaller AREA) then:

i. it would be necessary to include an additional independent variable given by AREA*SHOPPING and we would expect its coefficient to be negative.
ii. it would be necessary to include an additional independent variable given by AREA*SHOPPING and we would expect its coefficient to be positive.
iii. it would be necessary to include an additional independent variable given by the square of AREA (given by AREA2) and we would expect its coefficient to be positive.
iv. None of the previous answers is correct.

(c) Select the most appropriate sentence regarding all the stores owned by the retail company:

i. Stores with more floor area (with all other characteristics held constant) on average sell more because the coefficient on AREA is positive.
ii. Stores with more floor area sell more because the coefficient on AREA is positive.

iii. There is no evidence that stores with more floor area (with all other characteristics held constant) on average sell more for a significance level $\alpha = 5\%$.

iv. There is evidence that stores with more floor area (with all other characteristics held constant) on average sell more for a significance level $\alpha = 5\%$.

(d) It is estimated that if the number of competing stores in an urban area increases by 2, everything else held constant, then the store in that area will:

i. increase sales by one thousand euros

ii. decrease sales by one thousand euros

iii. decrease sales by 0.5 thousand euros

iv. most probably not have any impact on sales given that the p-value for that variable is practically zero.

(e) The value of $R^2$ in this regression means that:

i. the independent variables included in this model explain 94% of the variation in sales across the 145 stores in this sample.

ii. the percentage of variation in sales that is not explained by the independent variables included in this model equals 94%.

iii. the independent variables included in this model explain 0.94% of the variation in sales across the 145 stores in this sample.

iv. a 1% increase in the value of all the independent variables included in this model (except for the intercept) leads to an expected variation in sales of 94%.

(f) It is estimated that, everything else held constant, the stores that sell more on average are:

i. more than 5 km away from the city center and located outside a shopping mall.

ii. more than 5 km away from the city center and located inside a shopping mall.

iii. within 1 km from the city center and located outside a shopping mall.

iv. within 1 km from the city center and located inside a shopping mall.

(g) Select the best statement to explain what would happen to the value of $R^2$ if the regression model was re-estimated excluding the AREA variable?

i. would not increase.

ii. would not decrease.

iii. could increase, remain the same or decrease.

iv. would remain the same.

(h) Consider the following statement regarding the stores owned by the retail company: "if urban income increases by one thousand euros then store sales also increase by one thousand euros, assuming everything else held constant". Which of the following comments to this statement is correct?

i. This statement is clearly incorrect since the estimated coefficient of INCOME is less than one.

ii. This statement is correct since the p-value of INCOME is below 5%.

iii. This statement is incorrect since the p-value of INCOME is below 5%.

iv. For the usual confidence level of 95%, the available evidence is consistent with that statement.

(i) One variable that was not included in the regression model because of lack of data was the distance of each store to the nearest car parking lot. Usually shopping malls have nearby parking lots. What can be said about the estimated coefficient of the variable SHOPPING appearing in the regression above? Justify your answer.

(j) Provide some suggestions that in your opinion could lead to an improved regression model to be used to predict store sales. Justify your proposals.

# 7 Introduction to categorical data analysis

1. A professor is planning to use a new book for a financial accounting course and is considering three possibilities: Financial Accounting Made Easy, Financial Accounting Without Tears, and Financial Accounting for Profit and Pleasure. The professor contacted a random sample of 60 students who had already taken this course and asked each to review the three books, indicating a first preference. The results obtained are shown in the following table. Test the null hypothesis that for this population their first preferences are evenly distributed over the three books.

| Book | Made Easy | Without Tears | Profit and Pleasure |
|---|---|---|---|
| Number of first preferences | 17 | 25 | 18 |

2. (Newbold, p. 608) A charity solicits donations by telephone. It has been found that 60% of all calls result in a refusal to donate; 30% result in a request for more information through the mail, with a promise to at least consider donating; and 10% generate an immediate credit-card donation. For a random sample of 100 calls made in the current week, 65 result in a refusal to donate, 31 result in a request for more information through the mail, and 4 generate an immediate credit-card donation. Test at the 10% level the null hypothesis that the usual patter of outcomes is being followed in the current week.

3. A restaurant keeps record of the orders of all its customers. In the *Excel* file *Restaurant.xlsx* you can find a summary of the orders of a random sample of 40 customers.

   (a) Construct a contingency table relating entrée and dessert orders for this sample of customers. (Note: you can use the tool *Insert Pivot Table* from *Excel.*

   (b) Test, at a 5% signifincance level, whether there is enough evidence to support the claim that customers order meat, fish or vegetarian entrées with different probabilities.

   (c) Test, at a 5% signifincance level, whether there is evidence of some relationship between the entrée and the dessert orders.

4. Sally Smith is a long-time political campaign manager from Chicago. In the primary election there are four candidates, A, B, C and D. She wishes to determine if voter preference is different over the four major districts. A random sample survey results in the candidate preference frequencies by district and is shown in the following contingency table. Perform an appropriate statistical test to determine if candidate preference is related to the district.

| District | A | B | C | D | Total |
|---|---|---|---|---|---|
| 1 | 52 | 34 | 80 | 34 | 200 |
| 2 | 33 | 15 | 78 | 24 | 150 |
| 3 | 66 | 54 | 141 | 39 | 300 |

5. Data from supermarket scanners are frequently used by marketing researchers to understand the purchasing patterns and preferences of consumers. Researchers frequently study the purchases of a sample of consumers, called a scanner panel. When shopping, these consumers present a magnetic identification card that allows their purchase data to be identified and stored for subsequent statistical treatment. The table below reports peanut butter brand preferences obtained from data collected by a marketing research company for a consumer panel with 200 consumers in the city of "New Lisbon" during a 1-month period.

| Brand | No. costumers |
|---|---|
| Jeffy | 60 |
| Petty | 80 |
| Sky | 60 |
| Total | 200 |

(a) Do the data provide enough evidence to conclude that the preferences for the three brands are not equally distributed? Use a statistical hypothesis test approach to answer this question. Present your calculations.

(b) To better understand differences in preferences between men and women, the figures in the previous table are now discriminated between men and women in the table below.

| | No. costumers | |
|---|---|---|
| Brand | Men | Women |
| Jeffy | 30 | 30 |
| Petty | 30 | 50 |
| Sky | 20 | 40 |
| Total | 80 | 120 |

Do the data provide enough evidence to conclude that men and women have different brand preferences? Use a statistical hypothesis test approach to answer this question. Present your calculations.

(c) Are the conclusions derived in your answers to the previous questions, based on the panel of 200 consumers, applicable to the general population of consumers in "New Lisbon"? Discuss how confident you are with the results obtained from the statistical tests. What questions do you think are relevant to ask the marketing research company regarding how the data was obtained that may clarify any doubts you may have?

6. For the next academic year, the UVP University will require all its students to purchase their own laptop to use in class. To assess the reaction of UVP students to this new policy, a sample of students of the university was surveyed. To the question "What is your opinion about the proposed measure?" each student could answer "Agree", "Disagree", or "Indifferent". The results, in percentage, are presented in the following table which also distinguishes between undergraduate and master students.

| | Program | |
|---|---|---|
| Opinion | Undergraduate | Master |
| Agree | 20% | 30% |
| Disagree | 10% | 20% |
| Indifferent | 10% | 10% |

The results in the table were obtained for a sample of size 100. According to a student representative, for all the population of students of UVP, student's opinion on this matter is independent of the program of study. Perform a statistical test of this hypothesis using a significance level of 5%. What can you conclude?

7. What is the best style of business management? There are several studies about whether managers can get better results if they are more oriented towards executing tasks or more oriented towards people management. A recent study, based on a random sample of 150 companies, evaluated the predominant management style in each company and also the company performance. The results obtained are summarized in the following table.

| | Performance | |
|---|---|---|
| Management Style | Negative | Positive |
| Task Oriented | 40 | 20 |
| People Oriented | 20 | 20 |
| Balanced | 20 | 30 |

Given the results above, can one conclude that performance is not related to management style in the population of all companies (using a significance level of 5%)?

8. One of the most important issues in public debate today in various countries concerns the economic policy that governments must follow to effectively tackle economic crises. In particular, should the government cut public expenditures, raise taxes, or let the fiscal deficit increase? Suppose a random sample of 200 potential voters was asked about which of these three policies was preferable, and also about their political inclination (possible answers were left or right). The results obtained are summarized in the following table.

| | Political Inclination | |
|---|---|---|
| Opinion | Left | Right |
| Cut expenditures | 30 | 40 |
| Raise taxes | 10 | 20 |
| Let the deficit increase | 60 | 40 |

   (a) Before this survey was conducted, a journalist was convinced that the opinions in the population was divided as follows: 50% in favour of letting the deficit increase, 25% in favour of cutting expenditures, and 25% in favour of increasing taxes. With the available data, check the validity of this belief by using a statistical hypotheses test with a significance level of 5%. What can you conclude?

   (b) Do the results of the survey suggest that there is a relationship between the opinion about the economic policy to follow and the political inclination? To answer this question, conduct a statistical hypothesis test with a significance level of 5%. What can you conclude?

9. (Newbold, p. 619) How do customers first hear about a new product? A random sample of 200 users of a new product was surveyed to determine the answer to this question. Other demographic data such as age were also collected. The respondents included 50 people under the age of 21 and 90 people between the ages of 21 and 35; the remainder was over 35 years of age. Of those under 21, 60% heard about the product from a friend, and the remainder saw an advertisement in the local paper. One-third of the people in the age category from 21 to 35 saw the advertisement in the local paper. The other two-thirds heard about it from a friend. Of those over 35, only 30% heard about it from a friend, while the remainder saw the local newspaper advertisement. Set up the contingency table for the variables age and method of learning about the product. Is there an association between the consumer's age and the method by which the customer heard about the new product?

10. (Newbold, p. 619) Following a presidential debate, people were asked how they might vote in the forthcoming election. Is there any association between one's gender and choice of presidential candidate?

| | Gender | |
|---|---|---|
| Candidate preference | Male | Female |
| Candidate A | 150 | 130 |
| Candidate B | 100 | 120 |

# 8 Solutions of the proposed exercises

**Chapter 1**

1. $\mu = p$, $\sigma^2 = p(1-p)$
2. (a) 30%
   (b) 33.3%
3. (a) $\mu_X = 1.39$
   (b) $\mu_Y = 1.59$
   (d) 0.3399
   (e) No
4. $\sigma^2_{X_1} = \sigma^2_{X_2} = 0.6$, $Cov(X_1, X_2) = 0.4$, $\rho_{X_1,X_2} = 2/3$, $\sigma^2_{X_1+X_2} = 2$
5. (c) 0.75
6. $\mu = \frac{a+b}{2}$, $\sigma^2 = \frac{(b-a)^2}{12}$
7. (c) 0.25
   (d) 0.25
8. $\mu = 55000$, $\sigma = 12000$
9. 26.4; $\sigma_P = 1$
10. $\mu = 600$; $\sigma = 33.54$
12. (a) $\mu = 2850$, $\sigma^2 = 992500$
    (b) $\mu = 2850$; $\sigma^2 = 332500$
13. (a) They are the same.
    (b) It's best if they both supervise.
14. (a) They are the same.
    (b) Option B
15. (a) $\mu = 1400$
    (b) $\sigma = 482.183$
    (c) $\mu = 600$; $\sigma = 150$
    (d) i. $15X_B$
        ii. $13.5X_A + 1.5X_B$
        iii. $13.5X_A + 1.5X_B$
        iv. $10X_A + 5X_B$
16. (a) 0.8849
    (b) 0.0918
    (c) 0.9554
    (d) 0.8413
    (e) 0.0233
    (f) 0.8403
    (g) 0.1141
17. (a) 0.52
    (b) -0.67
    (c) 0.84
    (d) -0.25
18. (a) 0.1056
    (b) 0.9031
    (c) 0.7357
    (d) 56.72
    (e) $[49.52, 50.48]$
19. (a) 0.9772
    (b) 0.3674
    (c) 0.0062
    (d) 92.8
    (e) $[70, 90]$
20. (a) 0.2266
    (b) 0.2266
    (d) 100.6
21. (a) 0.0228
    (b) 0.0918
    (c) 0.3707
    (d) 7.568
22. (a) 0.5
    (b) 0.1587
    (c) 0.1586
    (d) 41.775
23. (a) 0.0013
    (b) 13.55%
24. (a) 0.4013
    (b) 4M euros
    (c) 16
    (d) 10.579
25. (a) 0.1587
    (b) 50
    (c) 180
    (d) -1
26. (a) 0.8413
    (b) 0.5
    (c) The second option is better
    (d) No
27. (a) 0.5793
    (b) i. 10
        ii. 10
        iii. $N(\mu = 10, \sigma^2 = 10^2)$
        iv. 0.1587
    (c) i. 10
        ii. $\sqrt{200(1-\rho)}$
        iii. $N(\mu = 10, \sigma^2 = 200(1-\rho))$
        iv. $N(\mu = 10, \sigma^2 = 0)$
28. (a) 0.0668
    (b) 37
    (c) i. 106
        ii. 1

**Chapter 2**

5. No
6. No
8. (a) 0.0796
   (b) 0.0796
   (c) 50
   (d) $\frac{10}{\sqrt{30}}$

(e) 0.4176
(f) 53.578
(g) Because they are independent
(h) It would not affect the answers

9. (a) 0.2586
(b) 0.6826
(c) 0.733
(d) (b) increase; (c) decrease
(e) Yes
(f) No

10. (a) No

11. (a) $\mu = 39.987$; $\sigma^2 = 101.541$; $\sigma = 10.077$; No; No
(b) $\mu = 39.987$; $\sigma^2 = 101.541$
(c) No

12. (a) 0.1587
(b) 108250 Km
(c) i. 100000
ii. 2500
iii. 0.9772
(d) i. Stays the same
ii. Increases
iii. Decreases

13. $\frac{\sigma_W^2}{100}$

14. $\frac{1}{4}$

15. (a) 100; $\frac{81}{25}$
(b) 0.1335
(c) 0.5788
(d) 0.7967

16. (a) 100; 30
(b) 0.0505
(c) 0.7337
(d) 0.8997

17. (a) i. 0.3085
ii. 0.1587
iii. 0.0228

18. (a) 0.9772
(b) 0.5762
(c) 0.3108
(d) \$214000 to \$216000

19. (a) 20
(b) 0.3085
(c) 0.9332
(d) Decrease; decrease; increase

20. (a) 0.3085
(b) 0.2266
(c) 0.3174
(d) Lower

21. (a) 0.264
(b) 0.2048
(c) 0.2304

22. (a) 68
(b) Smaller
(c) Larger

23. (a) 0.2593
(b) 0.0002

24. (a) 0.0004
(b) 25

26. 0.1587

27. (a) 0.2
(b) 0.0009
(c) 0.0298
(d) 0.0465

28. (a) 0.0285
(b) 0.0025
(c) 0.6111
(d) 0.41 to 0.43

29. (a) 0.0228
(b) 0.6826

30. 0.05

31. 1068

32. (a) 0.05
(b) 0.065
(c) 0.041

33. $\approx 0$

**Chapter 3**

1. (a) 2.57
(b) 0.0512
(c) 0.00512
(d) 0.7

2. (b) $\hat{\mu_1}$
(c) $\frac{5}{4}$; $\frac{10}{9}$

3. (a) iv.
(b) iii.

4. (a) $\mu$
(b) 9
(c) $N(\mu, 9)$

5. (a) i. $b = 1 - a \vee c = 0$
ii. $a = \frac{4}{5}$; $b = \frac{1}{5}$; $c = 0$
(b) $a = \frac{2}{3}$; $b = \frac{1}{3}$; $c = 0$

7. (a) They are all the same
(b) The third alternative

8. (a) They are both the same
(b) The second estimator

**Chapter 4**

1. (a) $]40.2; 59.8[$
(b) $]81.565; 88.435[$
(c) $]506.265; 513.735[$

2. (a) $]541.424; 578.576[$
(b) $]156.28; 163.72[$
(c) $]49.947; 66.053[$

3. (a) 13.918
   (b) 19.007
   (c) 3.493
   (d) 7.541
4. (a) $]174.076; 201.724[$
   (b) 96%
5. (a) $]6.804; 7.196[$
   (b) $]6.587; 7.413[$
   (c) iv
6. (a) $]3.993; 4.147[$
   (b) Narrower
   (c) Narrower
   (d) Wider
7. 64.24%
8. (a) $]68.068; 81.332[$
   (b) Wider
9. 68
10. (a) $]23.9345; 26.0655[$
    (b) 98%
11. (a) $]36.693; 39.307[$
    (b) It would increase
    (c) i. 90%
        ii. 13
12. (c)
14. $]29.208; 40.792[$
15. 29.125
16. (a) $N(\mu, 9)$
    (b) $]14.12; 25.88[$
17. (a) Below
    (b) $\bar{x} \pm z_{\alpha/2}\sqrt{\frac{(n+1)\sigma^2}{2n}}$
18. (a) $]0.1216; 0.2784[$
    (b) 0.098
19. (a) $]0.7216; 0.8784[$
    (b) 40%
    (c) 9604
20. $]0.0853; 0.1147[$
21. $]0.2026; 0.2974[$
22. $]0.585; 0.826[$
23. 84%
24. (a) $]0.909; 0.977[$
    (b) $]0.01; 0.07[$
25. (a) 1068
    (b) 385
26. (a) 664
    (b) 271
27. 752
28. (a) 1537
    (b) i. $]0.616; 0.884[$
        iii. $]-11.107; -8.893[$
        iv. $]0.057; 0.443[$
29. $]-0.087; 1.865[$
30. $]1296.85; 1703.15[$
31. $]27.064; 47.535[$
32. $]2.08; 9.92[$
33. $]-13.583; -4.417[$
34. $]-0.285; 0.145[$
35. $]0.031; 0.149[$
36. $\hat{p}_N - \hat{p}_O = 0.05$; $]-0.068; 0.1682[$
37. $]-0.096; 0.296[$
38. 777
39. $]-0.006; 0.062[$
40. $]-0.031; 0.082[$

**Chapter 5**

1. (b)
2. (c)
3. (a) $H_0 : \mu \leq 2000$ vs $H_1 : \mu > 2000$
   (b) Yes
   (c) The decision would be different
   (d) The decision would be different
4. (a) $H_0 : \mu \leq 50$
   (b) $H_0 : \mu > 50$
   (c) $\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \cap N(0,1)$
   (d) Reject $H_0$ if $\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} > 1.96$
   (e) i. 2.83
       ii. Reject $H_0$
       iii. $\approx 0.01$
       iv. $\alpha < 1\%$
   (f) i. 3
       ii. Reject $H_0$
       iii. 0.01
       iv. $\alpha < 1\%$
5. Do not reject $H_0$
6. (a) Reject $H_0$
   (b) Reject $H_0$
   (c) $\approx 0$; 0.02
7. Reject $H_0$
8. Reject $H_0$
9. No
10. (a) Do not reject $H_0$
    (b) 0.0808
    (c) Higher
11. (a) Reject $H_0$
    (b) Reject $H_0$
    (c) Do not reject $H_0$
    (d) Do not reject $H_0$
12. 0.0008
13. (a) Do not reject $H_0$
    (b) Yes
14. Do not reject $H_0$
15. $\approx 0$

16. Reject $H_0$
17. (a) Do not reject $H_0$
    (b) 29
18. (a) 0.29
    (b) 0.30
    (c) 0.28
    (d) 0.277
19. Reject $H_0$
20. 0.5222
21. 0.6966
22. (a) Do not reject $H_0$
    (b) 0.1
23. Reject $H_0$
24. Reject $H_0$
25. (a) Reject $H_0$
    (b) No
26. (a) $]0.031; 0.149[$
    (b) Reject $H_0$
27. Reject $H_0$
28. (a) Rejct $H_0$
    (b) 298
    (c) Reject $H_0$
    (d) No

**Chapter 6**

1. (b) 0.65
   (c) 4.4
   (d) $\hat{y} = 0.65x + 4.4$
2. (b) -45
   (c) -18
   (d) 194
   (e) 68
3. (b) -18.217
   (c) 268.7
   (d) 141.17
4. (b) Yes
   (c) $b_1 = 13.952; b_0 = -27.67$
5. (b) $b_1 = 3.695; b_0 = -3.569$
6. (a) Yes
   (b) $\hat{y} = 7.606x + 58.964$
7. (c) ii. 35.346
   (e) i) 278.4; ii) 0.9777
8. (a) $X$ - gross revenue; $Y$ - DVD revenue
   (b) ii. $b_0 = 10.473; b_1 = 0.384$
       vi. 39.266M$
9. (a) $\hat{Y} = 53.702 - 0.499B + 3.242O - 0.001T$
   (i) $CI_{90\%}(\beta_0) = [-43.319; 152.723]$;
   $CI_{90\%}(\beta_1) = [-0.831; -0.167]$;
   $CI_{90\%}(\beta_2) = [2.754; 3.729]$;
   $CI_{90\%}(\beta_3) = [-0.033; 0.031]$
   (j) $\hat{Y} = 50.071 - 0.499B + 3.230O$; Yes
   (k) $\hat{y} = 32.437$
13. (a) v. Reject $H_0$
14. (a) ii.
    (b) ii.
    (c) i.
    (d) iii.
    (e) ii.
    (f) ii.
    (g) Do not reject $H_0$
    (h) i.
15. (a) ii.
    (b) i.
    (c) ii.
    (d) ii.
    (e) iii.
    (f) i.
    (g) iii.
16. (a) $CI_{95\%}(\beta_1) = [0.683; 1.088]$
    (b) $CI_{90\%}(\beta_4) = [-3.034; -0.715]$
    (c) ii.
    (d) ii.
    (e) i.
    (f) ii.
    (g) i.
    (h) i.
    (i) iii.
    (j) Include the variable EXPERINCE*TRAINING
    (k) Do not reject $H_0$
17. (a) i.
    (b) i.
    (c) iii.
    (d) ii.
    (e) i.
    (f) iv.
    (g) i.
    (h) iv.
    (i) It is over-estimated

**Chapter 7**

1. Do not reject $H_0$
2. Do not reject $H_0$
3. (b) Do not reject $H_0$
   (c) Do not reject $H_0$
4. Do not reject $H_0$
5. (a) Do not reject $H_0$
   (b) Do not reject $H_0$
   (c) No
6. Do not reject $H_0$
7. Reject $H_0$
8. (a) Reject $H_0$
   (b) Reject $H_0$
9. Reject $H_0$
10. Do not reject $H_0$